

EXHIBIT A

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09:40:24 1 If you increase the bandwidth, you may let some more
09:40:27 2 frequencies through, and what that ends up giving you, on
09:40:31 3 a cell phone using this filter, is a clear signal. Like,
09:40:38 4 the voice sounds better or you can stream videos if the
09:40:42 5 bandwidth is larger, where you wouldn't be able to do it
09:40:45 6 with a narrow bandwidth.

09:40:52 7 **MR. LEMIEUX:** Excuse me, Your Honor. I just
09:40:53 8 wanted to clarify, this transcript is being sealed because
09:40:55 9 there is confidential information.

09:40:58 10 **THE COURT:** I think all material has covered
09:41:01 11 confidential information. All this information is
09:41:08 12 confidential, and there's no one in the room who was not
09:41:13 13 allowed to be in the room, I hope. I didn't see anybody.

09:41:19 14 **MR. LEMIEUX:** I didn't see anyone, Your Honor.
09:41:20 15 I just wanted to make sure the transcript is being
09:41:22 16 treated --

09:41:23 17 **THE COURT:** Absolutely. Thanks very much.

09:41:26 18 **BY MR. DeFOSSE:**

09:41:33 19 **Q.** All right.

09:41:36 20 **MR. DeFOSSE:** Can we go to PDX-6.196.

09:41:41 21 And Your Honor, may I approach?

09:41:44 22 **THE COURT:** You may.

09:41:44 23 **BY MR. DeFOSSE:**

09:42:03 24 **Q.** Dr. Shanfield, I've handed you also a hard copy of
09:42:06 25 the images that you see on the Slide 6.196 to assist you.

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09:42:14 1 If we look at this graph, can you explain to the jury
09:42:17 2 what we see on the left-hand side?

09:42:20 3 **A.** So what you see there is the structures I constructed
09:42:25 4 in the simulation. The top structure is the structure
09:42:30 5 that uses the '018 patent, and the bottom structure is
09:42:37 6 what the prior art was of structures at the time of that
09:42:43 7 patent. So there's thicknesses that are different in
09:42:48 8 each.

09:42:48 9 **Q.** Okay. And when you say that there's thicknesses that
09:42:52 10 are different in each, what do you mean?

09:42:54 11 **A.** So if you remember, the resonators that make up
09:42:59 12 filters have these two metal electrodes, metal on top,
09:43:03 13 metal on the bottom. And if you make the metal on the top
09:43:07 14 thinner than on the bottom, what the '018 patent says is
09:43:10 15 that you'll get better bandwidth.

09:43:13 16 So that's what I'm simulating here by doing thinner
09:43:19 17 on the top, thicker on the bottom, versus the same
09:43:22 18 thickness top and bottom.

09:43:23 19 **Q.** Okay. And just so we're clear, on the upper
09:43:27 20 right-hand side of the slide, in the blue box, that's the
09:43:32 21 Akoustis product that Qorvo is accusing of infringement;
09:43:35 22 is that right?

09:43:36 23 **A.** Upper left-hand side.

09:43:37 24 **Q.** Upper left-hand side. In the blue box?

09:43:39 25 **A.** Yes, that's correct.

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09:43:40 1 Q. And on the lower left-hand side of the green box,
09:43:44 2 that is a prior art product where the thickness of
09:43:48 3 electrodes is the same?

09:43:49 4 A. That's correct.

09:43:50 5 Q. So the difference between the lower box and the upper
09:43:52 6 box is, in the lower box, the electrodes are the same
09:43:55 7 thickness. In the upper box, in the Akoustis product, the
09:43:59 8 electrodes have different thicknesses?

09:44:01 9 A. That's correct.

09:44:01 10 Q. And where did you get the structures for the
09:44:04 11 thicknesses of the electrodes, the information on that?

09:44:08 12 A. From a combination of Dr. Heinrich and Dr. Bravman.

09:44:11 13 Q. Who the jury will hear from --

09:44:13 14 A. Yes.

09:44:13 15 Q. -- later today?

09:44:14 16 Okay. So you received the Akoustis structures and
09:44:17 17 the prior art structures, and you ran a computer
09:44:21 18 simulation; is that right?

09:44:22 19 A. That's correct, yes.

09:44:23 20 Q. And what was the result of the computer simulation?

09:44:26 21 A. The result was that the '018 patent works. It does
09:44:30 22 increase the bandwidth, as taught in that patent.

09:44:36 23 Q. Okay. So did you see that the Akoustis product had
09:44:39 24 an increased filter bandwidth?

09:44:41 25 A. I did.

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09:44:41 1 Q. Can you explain to the jury how you can see that in
09:44:45 2 the computer simulation results on the right-hand side of
09:44:48 3 196?

09:44:49 4 A. Yes. If you look at what I've circled in red, what
09:44:54 5 you see is a green line, and it's -- what it's
09:44:59 6 representing is the energy leakage, the rate of energy
09:45:05 7 leakage of the prior art product. And as -- and that's
09:45:14 8 been plotted as a function of frequency in the range of
09:45:17 9 the circle. You see that green line goes zooming up.
09:45:23 10 That means that energy leakage is way up at the band edge.
09:45:28 11 And the blue line is the structure with the '018 details
09:45:34 12 included. And, there, the blue line stays low. So at the
09:45:40 13 band edge, the loss is low. And that ends up making a
09:45:47 14 bigger bandwidth. So that is how I got my result.

09:45:52 15 Q. Okay. So the blue line represents the Akoustis
09:45:55 16 product?

09:45:56 17 A. Yes.

09:45:56 18 Q. And by having the blue line below the green line in
09:46:00 19 the red circle, you can see that the bandwidth of the
09:46:03 20 Akoustis product has increased?

09:46:05 21 A. That's correct.

09:46:05 22 Q. I'd like to turn to the '755 patent now.

09:46:14 23 MR. DeFOSSE: And if we could pull up
09:46:16 24 PDX-6.199.

09:46:19 1 **BY MR. DeFOSSE:**

09:46:19 2 **Q.** You were also asked to perform computer simulations
09:46:22 3 with respect to this patent, correct?

09:46:24 4 **A.** Yes, that's right.

09:46:25 5 **Q.** And, again, your simulations only related to one part
09:46:29 6 of the patent claim?

09:46:30 7 **A.** That's right.

09:46:31 8 **Q.** And the other parts are going to be addressed by
09:46:33 9 Dr. Bravman; is that right?

09:46:34 10 **A.** Yes.

09:46:34 11 **Q.** And for your simulations, which aspect of the claim
09:46:39 12 was relevant?

09:46:42 13 **A.** Again, it was at the bottom of -- it's Claim 9. And
09:46:46 14 what it's talking about is an arrangement in the structure
09:46:53 15 of the resonators such that there's less loss from the
09:47:00 16 resonator.

09:47:01 17 **Q.** So you simulated whether the Akoustis product would
09:47:07 18 result in having less energy loss?

09:47:11 19 **A.** Yes. Energy leakage is the same thing.

09:47:14 20 **Q.** Okay.

09:47:14 21 **MR. DeFOSSE:** If we go to PDX-6.200.

09:47:19 22 **BY MR. DeFOSSE:**

09:47:19 23 **Q.** Does this slide show the computer simulation that you
09:47:22 24 ran?

09:47:23 25 **A.** Yes, it does.

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09:47:24 1 Q. And on the left-hand side, in the blue box, what is
09:47:28 2 that?

09:47:29 3 A. So the blue box, again, is practicing what the
09:47:34 4 details are of the '755 patent and represent the Akoustis
09:47:39 5 device. And the structure below is representative of a
09:47:46 6 device that does not practice '755 details.

09:47:53 7 Q. And what's in the green box?

09:47:57 8 A. That -- I'm sorry. That's the bottom one.

09:47:58 9 The green box is what the structure is that does not
09:48:02 10 practice the '755 patent.

09:48:04 11 Q. Okay. So the blue box is the Akoustis product. The
09:48:07 12 green box is the prior art?

09:48:08 13 A. Prior art, yes.

09:48:10 14 Q. Okay. And did you determine whether, with your
09:48:14 15 simulation, the Akoustis product has less energy leakage?

09:48:21 16 A. Yes, I did.

09:48:21 17 Q. Okay. And is that shown in the graph on the
09:48:23 18 right-hand side?

09:48:24 19 A. Yes, it is.

09:48:24 20 Q. How is that?

09:48:26 21 A. So now, here, you're interested in the passband, and
09:48:33 22 I circled the passband of the filter. So that's the --
09:48:34 23 when I call it a "passband," I mean the frequency around
09:48:37 24 5.6 gigahertz in this graph. And what you see is the
09:48:45 25 Akoustis structure with the blue line has less loss in

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09:48:50 1 this passband than the green line, which is the prior art
09:48:53 2 structure. The green line is above the blue line in that
09:48:59 3 frequency range.

09:49:04 4 **Q.** Okay. Dr. Shanfield. Thank you.

09:49:06 5 **MR. DeFOSSE:** I will pass the witness.

09:49:08 6 **THE COURT:** Cross-examination.

09:49:10 7 CROSS EXAMINATION

09:49:10 8 **BY MR. LEMIEUX:**

09:49:46 9 **Q.** Good morning, Dr. Shanfield. You'll have to excuse
09:49:49 10 me as I grab my materials right here. You covered a lot
09:49:49 11 of ground, so I have a lot of questions for you.

09:49:57 12 In your analysis of the various groups of trade
09:50:01 13 secrets, you looked at numerous e-mails and other
09:50:10 14 materials that were presented to you from counsel for
09:50:14 15 plaintiff, right?

09:50:14 16 **A.** What I did was take the material that had been
09:50:23 17 discovered on Akoustis' computers and narrowed down what I
09:50:30 18 wanted to look at and did it like a funnel arrangement
09:50:35 19 where I tried to narrow it down to the point where I
09:50:38 20 wouldn't have to spend three years looking at it.

09:50:41 21 **Q.** And you said you wanted to do this to try to figure
09:50:44 22 out whether or not Akoustis received some type of
09:50:48 23 head-start benefit from its use or access or exposure to
09:50:53 24 this Qorvo information; is that right?

09:50:56 25 **A.** Well, I was looking for, I think I had it in my

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09:50:59 1 slide, these characteristics of whether it was known or
09:51:06 2 would be confidential information, whether there was an
09:51:11 3 awareness that the people handling that information was
09:51:15 4 confidential, whether there was economic value. So I
09:51:21 5 don't mean to go through the whole list. But that's the
09:51:24 6 characteristics I was looking for when I looked at the
09:51:28 7 group of -- this huge pile of data.

09:51:31 8 **Q.** So your head-start calculation is based on what you
09:51:35 9 believe to be the time and expense that Akoustis avoided
09:51:39 10 by allegedly using Qorvo's information and materials has a
09:51:43 11 shortcut; is that right?

09:51:45 12 **A.** I estimated the time savings; I didn't go into
09:51:49 13 expense.

09:51:50 14 **Q.** And so, in fact, the GAN chart that you showed up
09:51:54 15 here earlier is all just an estimate of time involved in
09:51:58 16 doing so; is that right?

09:51:59 17 **A.** It's not a GAN chart.

09:52:02 18 **Q.** Sorry?

09:52:03 19 **A.** It looks like a GAN chart, but, actually, it's just,
09:52:05 20 when I say, "year 1, year 2," those are meant to be
09:52:12 21 arbitrary times. It doesn't mean it started in year 1
09:52:16 22 that Akoustis was founded, for example. It's -- year 1 is
09:52:20 23 the year that this -- my experience tells me a startup
09:52:30 24 company would save this amount of time in doing these
09:52:34 25 tasks first. So it just put in order all the tasks so I

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could make sure I accounted for how they were done in parallel and not overestimate time savings.

Q. So you divided them into year time periods on your chart -- Year 1, Year 2, Year 3, Year 4, Year 5 -- that's right? I saw that accurately, didn't I?

A. It's not divided. I should have called it "Year A, Year B, Year C, Year D." They're not -- they are in order in terms of how I would see a startup company executing these tasks.

Q. Okay. And in terms of the order that you would see them executing a task, I would assume that if something in your chart was further over to the left, it's because you believed that would have been a task they would have attended to earlier in their formation, the items you listed towards the right-hand side are items that you would consider they would have gotten to later in their attention; is that right?

A. No.

Q. So that the right to left -- or, excuse me, the left to right head start you've calculated, that doesn't actually mean that?

A. What it means is I am plotting the benefit, the savings in time the company realized, and I am putting approximately where that savings might have been realized, but it's somewhat arbitrary. I just wanted to place it in

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1 have to be at when the moment this company was founded.

2 The advantage could have been realized somewhere else in
3 the history of the company. I just wanted to make sure,
4 and I'm roughly accounting for how the sequence of
5 benefits took place, but I didn't try and predict the
6 history of the company itself.

7 **Q.** Now, you've mentioned several times, Dr. Shanfield,
8 that you based these determinations in terms of number of
9 months when they would occur on your experience running a
10 startup company; is that right?

11 **A.** That's partly what I did, yes. I also accounted for
12 what I knew, for example, Qorvo would do, what I saw in my
13 experience in corporate life too.

14 **Q.** Well, your last experience in running a startup
15 company was in 2003, wasn't it?

16 **A.** Yes, that's right.

17 **Q.** Okay. So you haven't run one for over 20 years?

18 **A.** Yes.

19 **Q.** And in determining the months that you were looking
20 at for each period of each activity, is that estimation
21 based on some published journal somewhere, some study, or
22 is this just a number that you literally have generated
23 yourself out of thin air?

24 **A.** In every case, I considered where I could find
25 information that would give me a reference point. In some

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10:01:08 1 cases, I thought about projects that I had done in
10:01:12 2 corporate life recently at Draper Labs maybe as few as two
10:01:17 3 years ago. Other cases, I talked to somebody at Qorvo to
10:01:24 4 see how long this took. In other cases, I looked back in
10:01:28 5 the records I had before we sold our company as to when
10:01:32 6 and what we did so that I remembered, and, you know, I had
10:01:37 7 some kind of sense of what made -- what would be fair and
10:01:42 8 what made sense in a startup company.

10:01:44 9 **Q.** Now, you made reference a moment ago to Draper Labs.
10:01:48 10 Draper Labs is not a manufacturing company, is it?

10:01:51 11 **A.** No. And so that was more in the development aspects
10:01:53 12 and the procedure aspects, because we do provide low --
10:01:58 13 low-rate manufacturing in some cases.

10:02:01 14 **Q.** So, for example, in terms of, you know, assembly,
10:02:03 15 packaging, and all that, that's not anything that Draper
10:02:08 16 Labs does, right?

10:02:09 17 **A.** Yeah. I think I used my experience in assembly and
10:02:12 18 packaging from -- I ran a 24/7 semiconductor company that
10:02:17 19 delivered hundreds of millions of parts a year.

10:02:20 20 **Q.** When was that, Dr. Shanfield?

10:02:22 21 **A.** That was in the '90s.

10:02:24 22 **Q.** In the '90s. And have there been any changes?

10:02:26 23 **A.** And my experience with all the requirements you need
10:02:29 24 to be able to do that manufacturing. So that's where I
10:02:32 25 looked when I thought about, well, what does it take to

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10:03:58 1 different demands on it, much more advanced in general,
10:04:03 2 much more expensive kind of technology.

10:04:06 3 **Q.** So taking you back in to your determination of how
10:04:09 4 you assigned a certain amount of months for each of the
10:04:12 5 activities that you looked at here, is that assessment of
10:04:17 6 months for each task, like I said, is that based on any
10:04:20 7 other peer-reviewed or trade journal or any other
10:04:25 8 published material other than your own private estimation
10:04:28 9 that you did for this case?

10:04:36 10 **A.** I think I did look in one case at a publication
10:04:43 11 that -- I'm trying to recall exactly what the subject
10:04:49 12 matter was. I remember. It was in setting up a
10:04:54 13 reliability facility. I had a pile of publications, and I
10:05:04 14 glanced through them and looked at what, both what we had
10:05:07 15 done at Raytheon, and also what people who had published
10:05:13 16 information about -- evaluating filter reliability had
10:05:21 17 done so that I got a perspective, I saw the named
10:05:24 18 companies. So I guess I did in that case look at actual
10:05:31 19 published information.

10:05:32 20 **Q.** So you believe that in that particular instance, you
10:05:35 21 may have seen something. If you had seen it, that would
10:05:37 22 have been included as a reference to your expert report,
10:05:42 23 correct, as something you relied on?

10:05:48 24 **A.** I don't recall whether it was or not, but I do recall
10:05:51 25 I spent a lot of time making sure I was coming up with a

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10:05:57 1 reasonable number based on the best information I could
10:05:59 2 find, whether it was published document. A lot of times
10:06:03 3 this stuff isn't published because companies don't want to
10:06:07 4 share what it actually took for them to develop their
10:06:09 5 reliability setup. So I had to kind of read in between
10:06:14 6 the lines in documents I read --

10:06:17 7 **Q.** So, again, Dr. Shanfield, my question to you --

10:06:19 8 **A.** -- (indiscernible).

10:06:19 9 (Reporter clarification.)

10:06:19 10 **BY MR. LEMIEUX:**

10:06:19 11 **Q.** I'm sorry. And I didn't mean to interrupt you.

10:06:25 12 That's actually my fault.

10:06:28 13 So, Dr. Shanfield, getting back to my question. If
10:06:30 14 you had actually relied on something, it would be
10:06:33 15 reflected in your expert report, correct, because you have
10:06:36 16 an obligation to disclose every publication you relied on
10:06:39 17 in rendering your opinions in this case?

10:06:42 18 **A.** I agree.

10:06:43 19 **Q.** So if there isn't such a reference in your expert
10:06:46 20 report, because there wasn't one, correct?

10:06:49 21 **A.** I don't recall. There were so many documents. I
10:06:52 22 just don't recall.

10:06:53 23 **Q.** Now, again, getting to this time period of how you
10:06:56 24 get it, it sounds to me like if there is no published
10:07:00 25 guides for you to look at to determine this, then each of

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10:07:02 1 these time periods is really just your guesstimate of what
10:07:06 2 you assigned to each of these tasks, both as to when they
10:07:09 3 would occur and how long it would take to make them
10:07:13 4 accomplished; isn't that correct?

10:07:15 5 **A.** I'm not sure what you mean by "guesstimates." Like I
10:07:18 6 said, dependent on the time period we're talking about,
10:07:23 7 and the activity, as to where I went. I even asked
10:07:27 8 colleagues in one case. So I looked for sources that made
10:07:33 9 me feel comfortable that I was estimating correctly in
10:07:37 10 every case.

10:07:38 11 **Q.** Okay. And the sources that you relied on were what
10:07:43 12 exactly?

10:07:45 13 **A.** Dependent on the time period we're talking about.
10:07:48 14 Like I said, in the case of the reliability benefit, the
10:07:55 15 head start. I actually talked to an old friend who had
10:08:00 16 set up a facility. I looked at published papers on
10:08:05 17 reliability of filter devices, and in some cases, there
10:08:09 18 was -- at least, reading between the lines some hint of
10:08:14 19 what it would take.

10:08:15 20 **Q.** But, again, Dr. Shanfield, if you had relied on a
10:08:19 21 conversation with an unnamed old friend, if that was the
10:08:22 22 basis for your opinion, you would have had to disclose
10:08:24 23 that in your expert report as something that you
10:08:27 24 considered in arriving at your opinion, right?

10:08:30 25 **A.** That's right.

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10:08:31 1 Q. And so if there is nothing in your report identifying
10:08:35 2 this unnamed old friend, then that was not something that
10:08:39 3 you relied on in coming to your opinion in this case, was
10:08:42 4 it?

10:08:44 5 A. That's right. I agree.

10:08:46 6 Q. And you mentioned, you said you thought you talked to
10:08:50 7 Qorvo. On which of these items in any of these groups
10:08:53 8 that you've identified did you talk to Qorvo to find out
10:08:56 9 how long it took them to develop the alleged trade secret
10:08:59 10 in this case?

10:09:01 11 A. I talked with people at Qorvo about all of these
10:09:05 12 items. I didn't necessarily think they had the only word
10:09:10 13 on it, but I reviewed them to make sure what they were
10:09:14 14 saying.

10:09:15 15 Q. And, again, if you had discussed any -- with any
10:09:19 16 particular Qorvo individual, any of these particular trade
10:09:22 17 secrets, that would have been listed in your report as
10:09:24 18 well, correct, as sources that you relied on?

10:09:27 19 A. Yes.

10:09:29 20 Q. And as you sit here today, can you identify the names
10:09:32 21 of any Qorvo individuals that you spoke to in arriving at
10:09:36 22 these estimates?

10:09:36 23 A. Sure. Dr. Aigner, I probably got him tired out
10:09:42 24 discussing this with him.

10:09:43 25 Q. Which categories did you discuss with Dr. Aigner?

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10:09:46 1 **A.** I talked about all of them. In some cases, he said
10:09:50 2 he didn't know. So, but I wanted to go through all of
10:09:53 3 them.

10:09:54 4 **Q.** And in situations where Dr. Aigner reported to you
10:09:56 5 that he didn't know, then how did you come up with your
10:10:00 6 estimate?

10:10:00 7 **A.** It depended. Maybe you want to pick a particular
10:10:05 8 time period and how I estimated it? I've given you a
10:10:11 9 couple of examples; I don't want to keep...

10:10:14 10 **Q.** Well, for example, one of the items you took was the
10:10:18 11 evaluation of 5.2 and 5.6 gigahertz products?

10:10:23 12 **A.** Yes.

10:10:23 13 **Q.** And those were products that Akoustis actually came
10:10:26 14 to market with before Qorvo, correct?

10:10:28 15 **A.** Yes.

10:10:30 16 **Q.** Because, in fact, Qorvo was actually looking at
10:10:32 17 Akoustis products to fill in gaps in its product line
10:10:35 18 because it had not devoted the R&D resources to come up
10:10:40 19 with products for the Wi-Fi space in those frequencies,
10:10:43 20 correct?

10:10:44 21 **A.** That's correct, yes.

10:10:44 22 **Q.** So you estimated quite a long time period there as to
10:10:48 23 the evaluation of something that Akoustis actually already
10:10:52 24 had products in and Qorvo was a latecomer to that area.
10:10:55 25 Why did you estimate the 22 months that you claim it would

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10:28:32 1 **BY MR. LEMIEUX:**

10:28:33 2 **Q.** Here, you've evaluated what you believe to be the
10:28:36 3 head-start advantages in various items in Group 5; is that
10:28:39 4 correct?

10:28:41 5 **A.** Yes, that's correct.

10:28:43 6 **Q.** And in your evaluation of the head start that was
10:28:48 7 allegedly provided here, did you review at all the library
10:28:52 8 of processes and development documents that were already
10:28:55 9 in place at the New York fab facility?

10:29:03 10 **A.** I don't recall, frankly. There's so many documents.
10:29:07 11 I have to go back in my report and check.

10:29:10 12 **Q.** I can tell you, Dr. Shanfield, you did not. At least
10:29:12 13 it was not listed in your report as items that you had
10:29:15 14 reviewed in arriving at your opinion.

10:29:17 15 **A.** Okay.

10:29:18 16 **Q.** So when you estimate the advantage of creating a
10:29:23 17 technology development process or product development
10:29:26 18 process, applicable testing procedures, parameters, you
10:29:32 19 had no idea of what was already actually in place at the
10:29:37 20 New York fab facility for these purposes, do you?

10:29:39 21 **A.** Well, actually, I did because I had seen it. But it
10:29:43 22 didn't matter. I didn't use that. What I'm aware of that
10:29:48 23 testing BAW filters is really specialized, and there's a
10:29:53 24 lot of detail to getting it right. And that is not
10:29:58 25 something that the MEMS fab had the ability of doing, in

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10:30:03 1 the information I had available.

10:30:04 2 **Q.** So you believe a fab has years of experience in
10:30:07 3 making MEMS devices, has no procedures in place for
10:30:10 4 testing the devices its making?

10:30:12 5 **A.** Remember, the testing that you have to do for MEMS
10:30:17 6 filter is RF testing at high frequency, much higher than
10:30:24 7 any MEMS normally runs. There's no MEMS resonator at that
10:30:27 8 frequency that's any good. No MEMS facility I've ever
10:30:31 9 seen has the kind of testing that -- you have to calibrate
10:30:37 10 that testing in a very elaborate way with S parameter
10:30:40 11 measurements. All of that is really specialized, and
10:30:44 12 particular to BAW filters.

10:30:45 13 So that was, my view, something that was not
10:30:49 14 established at the fab. And I took that into account.

10:30:53 15 **Q.** But you've listed --

10:30:55 16 **THE COURT:** It's 10:30, and, of course, we have
10:30:58 17 a ways to go. This is our 15-minute morning break. So,
10:31:02 18 remember, don't discuss the case among yourself. Don't
10:31:06 19 let anybody talk to you about it. Of course we will come
10:31:09 20 back and resume. Keep an open mind. Fifteen minutes.
10:31:13 21 Thanks very much.

10:31:16 22 (The jury exits the courtroom at 10:31 a.m.)

10:31:44 23 **THE COURT:** Everyone can be seated. The
10:31:46 24 witness will be able to step down for a 15-minute break.
10:31:49 25 Can't talk with anybody about answers given, so forth.

Shanfield - Cross

10:31:52 1 But this is our morning break.

10:31:56 2 Anything else before we take that break right
10:31:58 3 now?

10:32:00 4 **MR. LEMIEUX:** No, I don't think so, Your Honor.

10:32:01 5 **THE COURT:** Let's give everybody a chance to
10:32:03 6 have that break. We will see everybody -- I will see you
10:32:07 7 in about 13 minutes. Thank you very much.

10:32:59 8 (Whereupon, a recess was taken.)

10:46:49 9 **THE COURT:** You may be seated. We can have the
10:46:54 10 witness back on the stand, and the jury can come in.

10:47:30 11 (The jury enters the courtroom at 10:47 a.m.)

10:47:55 12 **THE COURT:** Everybody can be seated, and
10:47:57 13 counsel may proceed.

10:47:58 14 **BY MR. LEMIEUX:**

10:48:00 15 **Q.** Doctor, just before the break here, we were talking
10:48:04 16 about the operation of New York fab. I wanted to make
10:48:07 17 sure I understood your testimony you've never actually
10:48:09 18 reviewed the fab's library of processes controls
10:48:13 19 procedures, have you?

10:48:18 20 **A.** No, I don't think so.

10:48:21 21 **Q.** So your opinion in this case is rendered without
10:48:23 22 actually looking at the actual procedures and processes
10:48:27 23 that are used in their fabrication in the manufacturing
10:48:31 24 facility in New York, correct?

10:48:32 25 **A.** That's correct, but what that --

Shanfield - Cross

1 11:50:56 1 Q. So, again, packaging and assembly information used by
2 11:51:00 2 Qorvo isn't going to be of much use to Akoustis if that
3 11:51:05 3 function is handled by a third party for Akoustis; isn't
4 11:51:08 4 that right?

5 11:51:09 5 A. I disagree.

6 11:51:44 6 Q. All right. Dr. Shanfield, I'd like to just, then,
7 11:51:47 7 move on to your involvement in the assessment of potential
8 11:51:52 8 patent infringement in this case.

9 11:51:53 9 A. Sure.

10 11:51:54 10 Q. You were asked to conduct certain simulations for
11 11:52:00 11 purposes of other experts in this case to comment on; is
12 11:52:05 12 that right?

13 11:52:08 13 A. Yes. That's a good description, yes.

14 11:52:12 14 Q. And you performed those simulations, and those
15 11:52:16 15 simulations produced certain graphs that you've referred
16 11:52:21 16 to in your testimony here today; is that right?

17 11:52:23 17 A. Well, I performed dozens of simulations, but I am
18 11:52:28 18 showing the results from a few.

19 11:52:30 19 Q. And the simulations you ran, they didn't consistent
20 11:52:33 20 of consist of actually physical testing of Akoustis
21 11:52:37 21 devices, did they?

22 11:52:38 22 A. No.

23 11:52:38 23 Q. It's just computer modeling in an attempt to project
24 11:52:42 24 what actual testing would show, right?

25 11:52:44 25 A. Yes. That was, I think --

Shanfield - Cross

1 1:52:49 1 Q. And for the original --

1 1:52:49 2 A. -- you heard from testimony and witnesses that, in
1 1:52:53 3 fact, Dr. Aigner, the modeling is the way people design
1 1:52:57 4 these devices now. You can't build lots of different
1 1:53:03 5 variations. It just takes too long and it's too hard, and
1 1:53:06 6 the modeling has gotten so good in the BAW resonators and
1 1:53:12 7 filters that you end up very accurately predicting what
1 1:53:15 8 the real world will do. So modeling is the Number 1
1 1:53:19 9 approach, and the first approach you take in designing and
1 1:53:22 10 deciding what you are going to build.

1 1:53:24 11 Q. Okay. So while modeling is a potentially useful
1 1:53:28 12 device in designing figures, that's not the question I
1 1:53:30 13 asked you.

1 1:53:32 14 We were looking at, looking at actual finished
1 1:53:35 15 devices and whether you did any physical testing of the
1 1:53:38 16 actual finished device?

1 1:53:41 17 A. I didn't do any. It was cross-sections taken of
1 1:53:45 18 actual physical devices.

1 1:53:47 19 Q. So, again, what you did was you tried to create a
1 1:53:51 20 computer model of what it would predict the actual device
1 1:53:55 21 would do in operation, correct?

1 1:53:58 22 A. No. What I did was, I took the data that was
1 1:54:02 23 supplied to me from scanning electron microscope images
1 1:54:07 24 and measurements of the actual device, and I took those
1 1:54:10 25 dimensions. I put them in a model that I know gives

Shanfield - Cross

1 1:54:14 1 accurate results, have long experience with it, and I
1 1:54:19 2 tested that model. I made dozens of simulations to make
1 1:54:24 3 sure the model behaved properly as expected. And then I
1 1:54:29 4 did the simulations I was asked to do.

1 1:54:34 5 Q. And simulations, as we heard from both Dr. Aigner and
1 1:54:37 6 Dr. Fattinger when they were on the witness stand, are
1 1:54:40 7 very complicated, aren't they?

1 1:54:42 8 A. Yes.

1 1:54:42 9 Q. And they consist of multiple variables that have to
1 1:54:45 10 be considered in order to produce a useful prediction;
1 1:54:49 11 isn't that right?

1 1:54:50 12 A. That's correct. Takes a lot of work to get it to
1 1:54:53 13 work, correct.

1 1:54:53 14 Q. And --

1 1:54:53 15 A. It's got to be checked a lot.

1 1:54:56 16 Q. And if there is a mistake if one of those variables,
1 1:54:58 17 it's going to end up producing a result that's incorrect
1 1:55:01 18 or not useful, right?

1 1:55:03 19 A. Yes. A mistake is very unlikely, because I did the
1 1:55:06 20 same as all people who do modeling a lot do, and I tested
1 1:55:12 21 the model to make sure, in the range of parameters I was
1 1:55:15 22 looking at, that it behave the way I know the physical
1 1:55:19 23 world behaves. So that you could check if there's
1 1:55:21 24 something wrong, you will see it in one of its
1 1:55:24 25 predictions.

Shanfield - Cross

1 1:55:26 1 Q. Now, you produced an expert report that contained --
1 1:55:29 2 well, let me step back for a second. So the results of
1 1:55:32 3 the computer modeling that you do, produced the graphs
1 1:55:35 4 that we saw up on the screen earlier this morning?

1 1:55:39 5 A. Yes.

1 1:55:39 6 Q. And in your original expert report, you produced
1 1:55:43 7 certain graphs that you believed depicted the actual
1 1:55:46 8 operation of the Akoustis devices, correct?

1 1:55:50 9 A. It predicted the performance of, in some cases,
1 1:55:54 10 Akoustis devices; in some cases what was the state of the
1 1:55:58 11 art at the time.

1 1:56:00 12 Q. And you reviewed your expert report, didn't you, to
1 1:56:03 13 make sure that it was accurate before it was provided to
1 1:56:08 14 counsel for Akoustis?

1 1:56:09 15 A. Yes.

1 1:56:10 16 Q. And, in fact, didn't it turn out that the graphs that
1 1:56:14 17 you provided in your original expert report were
1 1:56:17 18 incorrect?

1 1:56:19 19 A. I made a mistake of pasting the wrong graph into one
1 1:56:23 20 page, and the simulation itself that I pasted in was
1 1:56:29 21 correct, but it was not the one that I meant to refer to
1 1:56:33 22 in my writing. So I did, indeed.

1 1:56:36 23 Q. And even though that report was produced at towards
1 1:56:40 24 the end of November of 2023, you didn't notice that
1 1:56:46 25 mistake for several months; isn't that true?

Shanfield - Cross

1 1:56:50 1 **A.** That's true. I have a full-time job. I have lots of
1 1:56:53 2 other things to worry about.

1 1:56:54 3 **Q.** And, in fact, you didn't notice that mistake until it
1 1:56:57 4 was pointed out in the rebuttal reports submitted by
1 1:57:00 5 Akoustis' expert; isn't that true?

1 1:57:02 6 **A.** No, that's not true.

1 1:57:03 7 **Q.** And isn't it also true that you weren't aware of this
1 1:57:06 8 until Dr. Bravman, who we'll hear from later, pointed out
1 1:57:10 9 the mistake to you in your expert report?

1 1:57:12 10 **A.** That's not true.

1 1:57:14 11 **Q.** So if Dr. Bravman testifies that he had to point this
1 1:57:20 12 mistake out to you, he would be incorrect in that
1 1:57:23 13 testimony?

1 1:57:24 14 **A.** Dr. Bravman pointed to that plot and found it wasn't
1 1:57:27 15 consistent with the language I was referring to, and I
1 1:57:33 16 looked at it and immediately knew what I had done, and I
1 1:57:37 17 pasted in the right plot.

1 1:57:41 18 **Q.** And so you had to change your report, didn't you?
1 1:57:43 19 Had to supplement it to put in different graphs than what
1 1:57:47 20 you had included in your original report?

1 1:57:50 21 **A.** I pasted in the wrong graph. I took that out and put
1 1:57:53 22 the right one in. The simulations were all correct.
1 1:57:56 23 There were no errors.

1 1:58:10 24 **MR. LEMIEUX:** No further questions for this
1 1:58:11 25 witness, Your Honor.

Heinrich - Direct

15:05:45 1 in size. You see the schematic on the left side there.

15:05:49 2 And I had to cut the packaging open to get the real
15:05:54 3 device out, and I had a little micro saw that is a very
15:05:57 4 small, very fine circular saw here that turns very slowly.
15:06:02 5 And in about 15, 20 minutes I can get the real device out
15:06:07 6 of it.

15:06:08 7 **MR. TIGAN:** Let's turn to the next slide, 9.6.

15:06:13 8 **BY MR. TIGAN:**

15:06:13 9 **Q.** Here, you have Step Number 2. What does that entail?

15:06:16 10 **A.** So what you see on the right side is the system that
15:06:18 11 I used for sample preparation prior to sample preparation
15:06:20 12 imaging. This is a scanning electron microscope with some
15:06:27 13 students in front of it, not on the sample, obviously.
15:06:28 14 But on the left side, you see a scanning electromicrograph
15:06:33 15 or image of -- top view of one of the resonators, and that
15:06:37 16 was AKF1256 here.

15:06:43 17 **MR. TIGAN:** Okay. Let's go to the next slide.

15:06:45 18 **BY MR. TIGAN:**

15:06:45 19 **Q.** And I would like to ask you what Step 3 is.

15:06:48 20 **A.** Yes. So this scanning electron microscope is also
15:06:51 21 equipped with ion beam that I use to remove material from
15:06:56 22 the surfaces. What you see on these labeled micrograph on
15:06:59 23 the left side is this resonator we named LR6. You see
15:07:05 24 this little oval that is magnified there, and you see this
15:07:09 25 little slightly tilted line there. This is micro section,

Heinrich - Direct

15:07:13 1 but I'm still viewing it from top. And above these two
15:07:17 2 rectangles that you see there above and below, these are
15:07:20 3 the material that I removed from the sample during the
15:07:24 4 cross-section preparation.

15:07:26 5 Q. Okay.

15:07:27 6 MR. TIGAN: Let's go to the next slide, 9.8.

15:07:30 7 BY MR. TIGAN:

15:07:30 8 Q. And I believe this is Step 4. Can you walk us
15:07:33 9 through what you've put on here?

15:07:34 10 A. Yes. So there you are seeing me putting the sample
15:07:39 11 into my transmission electron microscope. And I analyzed
15:07:44 12 the cross-sections of these materials. You see -- on top
15:07:48 13 left, you see an image of whole cross-section that I
15:07:53 14 prepared. It looks a little bit like a piece of bacon,
15:07:56 15 but you have to imagine this piece of bacon being about
15:08:00 16 5,000 times smaller in every direction, so in thickness,
15:08:04 17 length, and width, right? And in the bottom view you see
15:08:08 18 a cross-section view of the same sample that is just at
15:08:12 19 higher magnification with some calibrated thickness
15:08:18 20 measurements that I did.

15:08:19 21 MR. TIGAN: And if we can remove that blow-out.

15:08:22 22 BY MR. TIGAN:

15:08:22 23 Q. I just want to confirm that, on the top left, is that
15:08:24 24 an entire resonator cross-section?

15:08:27 25 A. That's a cross-section across the entire resonator,

Heinrich - Cross

15:08:30 1 yes.

15:08:30 2 **Q.** And like I said, we won't go through all of these,
15:08:32 3 but just to confirm, did you perform these same steps for
15:08:35 4 all of the cross-sections that you go through in the
15:08:38 5 appendixes?

15:08:39 6 **A.** Yes.

15:08:40 7 **Q.** One last question for you. We're going to wrap up
15:08:42 8 quickly. Could you characterize the margin of error for
15:08:46 9 your measurements here?

15:08:48 10 **A.** Yes. My margin of error of the calibrations of the
15:08:50 11 microscope is about 1 to 2 percent.

15:08:55 12 **Q.** Okay. That's all the questions I have for now?

15:08:57 13 **MR. TIGAN:** I pass the witness, Your Honor.

15:08:58 14 **THE COURT:** Cross-examination.

15:08:59 15 CROSS EXAMINATION

15:09:01 16 **BY MS. SMITH:**

15:09:01 17 **Q.** Good afternoon, Dr. Heinrich. How are you?

15:09:04 18 **A.** Good afternoon. Thank you.

15:09:05 19 **Q.** You testified that you were provided with samples of
15:09:09 20 19 different Akoustis products, right?

15:09:12 21 **A.** Correct, yes.

15:09:12 22 **Q.** And you received multiple samples for each of the
15:09:16 23 products?

15:09:17 24 **A.** Correct.

15:09:17 25 **Q.** You tested only one sample of each product?

Bravman - Direct

15:53:16 1 **BY MR. MASTERS:**

15:53:16 2 **Q.** Very briefly, Dr. Bravman, did you also consider
15:53:19 3 Dr. Heinrich's cross-section?

15:53:20 4 **A.** Yes. As I indicated, I also considered and used the
15:53:23 5 information provided by Dr. Shanfield, and Dr. Heinrich.
15:53:26 6 You just saw this. This is just a summary slide.

15:53:29 7 Again, I did this kind of work personally starting as
15:53:33 8 a graduate student, and so I'm very familiar with this and
15:53:36 9 how these figures and information is produced.

15:53:40 10 **Q.** Okay.

15:53:40 11 **MR. MASTERS:** And, Mr. Faison, can we project
15:53:43 12 PDX-10.10.

15:53:45 13 **BY MR. MASTERS:**

15:53:45 14 **Q.** And did you also rely on Dr. Shanfield's simulations?

15:53:50 15 **A.** Yes. At the end of Dr. Shanfield's testimony here
15:53:53 16 today, he described his work in doing finite element
15:53:58 17 modeling. This is a computer-based model involving, in
15:54:02 18 this case, mechanical interactions between various places
15:54:08 19 in a model structure. And I'll touch on this a little bit
15:54:14 20 more again, but you heard about this at somewhat length
15:54:17 21 this morning from Dr. Shanfield.

15:54:19 22 This is a standard modeling procedure. And so this
15:54:23 23 is something that I personally do not do, but I've used
15:54:26 24 throughout my career when done by others.

15:54:31 25 **Q.** Dr. Bravman, let's turn to the '018 patent.

Bravman - Cross

1 6 : 4 4 : 4 7 1 to understand why have you grouped the helper and the
1 6 : 4 4 : 5 0 2 shunt in the series all as part of the top electrode? If
1 6 : 4 4 : 5 3 3 they're not part of it, why wouldn't you have listed them
1 6 : 4 4 : 5 6 4 elsewhere?

1 6 : 4 4 : 5 6 5 **A.** Because they all have top and bottom electrodes.

1 6 : 4 4 : 5 9 6 **Q.** Okay. So, in this case, these helper and shunt
1 6 : 4 5 : 0 2 7 series, those are both showing larger sizes than the
1 6 : 4 5 : 0 5 8 bottom electrode layer?

1 6 : 4 5 : 0 7 9 **A.** Correct.

1 6 : 4 5 : 0 8 10 **Q.** Okay.

1 6 : 4 5 : 1 2 11 **MR. LEMIEUX:** If we could go to Page PDX-10.10.

1 6 : 4 5 : 2 0 12 **BY MR. LEMIEUX:**

1 6 : 4 5 : 2 0 13 **Q.** These are showing some of the diagrams or information
1 6 : 4 5 : 2 3 14 you received from Dr. Shanfield, right --

1 6 : 4 5 : 2 6 15 **A.** Yes.

1 6 : 4 5 : 2 6 16 **Q.** -- that assisted you in your analysis?

1 6 : 4 5 : 2 8 17 **A.** Yes.

1 6 : 4 5 : 2 9 18 **Q.** And on the graphs here, the larger graphs that are in
1 6 : 4 5 : 3 2 19 the middle, they say "lateral energy flux."

1 6 : 4 5 : 3 6 20 What's the unit of measurement that's along that
1 6 : 4 5 : 3 8 21 axis?

1 6 : 4 5 : 3 9 22 **A.** I don't know what the unit is. It's probably --
1 6 : 4 5 : 4 2 23 because this is a finite element model -- it was relative.
1 6 : 4 5 : 4 5 24 And so one would be a baseline. But, again, I didn't
1 6 : 4 5 : 4 9 25 produce these. And so Dr. Shanfield would have to answer

Bravman - Cross

1 6:51:06 1 Q. Okay. And so it's fair to say that your infringement
1 6:51:09 2 opinion, then, is based upon the simulations run by
1 6:51:14 3 Dr. Shanfield, in part, and in part by the testing done by
1 6:51:17 4 Dr. Heinrich?

1 6:51:18 5 A. Of course.

1 6:51:18 6 Q. Now, earlier in this case, Dr. Shanfield -- and we
1 6:51:26 7 talked about it a little bit in his testimony today --
1 6:51:29 8 produced an original expert report that had the kind of
1 6:51:33 9 graphic information we saw here a moment ago. But it
1 6:51:36 10 wasn't these graphs, was it?

1 6:51:38 11 A. I don't recall exactly what was in his report
1 6:51:40 12 vis-a-vis these we have seen today. But as you recounted,
1 6:51:44 13 or someone recounted earlier today, I found that there was
1 6:51:48 14 one that was erroneously included in his report. It just
1 6:51:51 15 didn't match his description.

1 6:51:53 16 Q. Right. And, in fact, you brought that, then, to
1 6:51:55 17 Dr. Shanfield's attention. And then he ran new testing or
1 6:51:59 18 produced a new graph in a supplemental report; is that
1 6:52:02 19 right?

1 6:52:02 20 A. He didn't do new testing. My understanding is he
1 6:52:07 21 simply corrected his cut-and-paste error.

1 6:52:10 22 Q. And so if your original opinion, however, on
1 6:52:13 23 infringement was based upon the material that had been
1 6:52:16 24 provided by Dr. Shanfield in his original report, wasn't
1 6:52:19 25 it?

Bravman - Cross

1 6:52:20 1 **A.** A study which yielded to me, as a worker of skill,
1 6:52:24 2 that this must be an error, only one. And I reported that
1 6:52:27 3 faithfully. And when it was corrected, it doesn't change
1 6:52:32 4 my opinion at all.

1 6:52:33 5 **Q.** Well, it wasn't corrected for several months, isn't
1 6:52:35 6 that true?

1 6:52:36 7 **A.** I'm not aware of the timing. I did my job and I
1 6:52:39 8 reported it.

1 6:52:40 9 **Q.** You submitted your original expert report in this
1 6:52:44 10 case in November of 2023; isn't that right?

1 6:52:47 11 **A.** Sounds about right.

1 6:52:47 12 **Q.** And Dr. Shanfield also produced his original report
1 6:52:50 13 in November of 2023?

1 6:52:52 14 **A.** If you represent that, I'll take that at your word.

1 6:52:55 15 **Q.** Well, because, in fact, your infringement opinion
1 6:52:58 16 relied upon analysis that he was doing. So that was the
1 6:53:01 17 report that you were relying on at the time, correct?

1 6:53:04 18 **A.** It obviously could not have been after my report.
1 6:53:07 19 That's correct.

1 6:53:07 20 **Q.** Okay. So we agree, then, that you were relying on
1 6:53:11 21 the original report submitted by Dr. Shanfield in this
1 6:53:14 22 case?

1 6:53:15 23 **A.** Yes.

1 6:53:15 24 **Q.** And so your infringement opinion, then, relied on
1 6:53:18 25 something that ultimately turned out to be not correct,

Bravman - Redirect

1 6 : 5 3 : 2 1 1 which you noticed months later; is that right?

1 6 : 5 3 : 2 4 2 **A.** It had one clearly cut-and-paste error in it. That's
1 6 : 5 3 : 2 8 3 right.

1 6 : 5 3 : 4 0 4 **Q.** I just want to check -- just bear with me for a
1 6 : 5 3 : 4 6 5 moment here. I may be done.

1 6 : 5 4 : 0 3 6 Now, you spoke for a few minutes at the end of your
1 6 : 5 4 : 0 7 7 testimony, Dr. Bravman, about what was called the '360
1 6 : 5 4 : 1 1 8 patent.

1 6 : 5 4 : 1 3 9 **A.** Yes.

1 6 : 5 4 : 1 3 10 **Q.** And you're not a patent licensing expert, are you?

1 6 : 5 4 : 1 5 11 **A.** No.

1 6 : 5 4 : 1 7 12 **Q.** So any opinions you may have as to the value is
1 6 : 5 4 : 2 1 13 something that's outside the area of your expertise and
1 6 : 5 4 : 2 4 14 outside what you've been asked to do in this case,
1 6 : 5 4 : 2 6 15 correct?

1 6 : 5 4 : 2 7 16 **A.** Monetary value, for sure outside my expertise.

1 6 : 5 4 : 3 1 17 **Q.** Okay.

1 6 : 5 4 : 3 1 18 **MR. LEMIEUX:** No further questions.

1 6 : 5 4 : 3 2 19 **THE COURT:** Redirect?

REDIRECT EXAMINATION

1 6 : 5 4 : 3 4 21 **BY MR. MASTERS:**

1 6 : 5 4 : 3 8 22 **Q.** Dr. Bravman, counsel asked you about the
1 6 : 5 4 : 4 4 23 cut-and-paste error in Dr. Shanfield's report.

1 6 : 5 4 : 4 7 24 **A.** Yes.

1 6 : 5 4 : 4 9 25 **Q.** Is it your understanding that there was both a

08:47:16 1 THE COURT: Okay. Yes, sir.

08:47:17 2 MR. ELKINS: So, Ms. Bennis will be testifying.

08:47:20 3 As Your Honor knows, we move to exclude her trade secrets
08:47:24 4 opinion under Daubert and Rule 702, for purposes of ensuring
08:47:32 5 that we have made a full, a full record for future purposes.

08:47:36 6 THE COURT: Sure.

08:47:37 7 MR. ELKINS: We just like to renew our motion,
08:47:39 8 Your Honor. We believe it's well taken, and that under
08:47:42 9 amended Rule 702, the Court has an obligation to exercise
08:47:47 10 its gatekeeping obligations by excluding those opinions,
08:47:52 11 understanding you have already ruled, Your Honor.

08:47:55 12 THE COURT: I'm just checking. The nice thing I
08:47:59 13 have two screens that actually work today.

08:48:01 14 MR. ELKINS: And that was the only thing that I
08:48:03 15 wanted to say before we began. Thank you.

08:48:07 16 THE COURT: Thank you. I think we're all set.
08:48:10 17 I hope the witness is ready today. Okay. I see the witness
08:48:15 18 is.

08:49:00 19 I think we're trying to send you a revised
08:50:19 20 verdict form. We'll get it to you fairly soon.

08:50:27 21 (Jury entering the courtroom at 8:50 a.m.)

08:50:56 22 THE COURT: All right. Everyone can be seated.
08:51:06 23 And that was some bad traffic I think coming in. Is that
08:51:10 24 right? I know somebody -- I thought someone reported that.
08:51:15 25 So I think we're all set. We're starting a little late

Bennis - direct

09:05:18 1 Q. How did you calculate the unjust enrichment damages
09:05:22 2 here?

09:05:23 3 A. This analysis started, the starting point for it was
09:05:29 4 Dr. Shanfield's analysis. So again, last week we heard that
09:05:34 5 all of the acquisition of confidential important information
09:05:40 6 allowed Akoustis to move into the market and do all the
09:05:44 7 business related things it was doing at least 55 months
09:05:47 8 early.

09:05:48 9 So it then turns to what benefit did that help
09:05:53 10 Akoustis achieve. Well, earning revenue. It earned revenue
09:05:56 11 earlier than it otherwise would have. So the analysis then,
09:06:00 12 the test for me, is how do you put a dollar figure on that?
09:06:03 13 How do you figure out the benefit of having that revenue
09:06:06 14 sooner than you otherwise would have?

09:06:08 15 Q. And how did you determine the monetary value of
09:06:11 16 Akoustis obtaining a 55-month head-start to get into the
09:06:15 17 market?

09:06:16 18 A. My analysis measured the monetary value by evaluating
09:06:21 19 the time value of having money earlier rather than later.

09:06:26 20 Q. What do you mean by the time value of money?

09:06:30 21 A. The time value of money is a very widely known
09:06:34 22 studied economic concept. To break it down more simply, the
09:06:41 23 concept is that having a dollar today, is always better than
09:06:45 24 having a dollar tomorrow. And the reason being, if I have
09:06:48 25 that dollar today, I can invest that dollar, I can earn

Bennis - direct

09:06:53 1 interest on that dollar. If I'm a company, I can use that
09:06:56 2 dollar to buy things, to build things, to make more money.
09:07:00 3 So there is power in having money sooner rather than later,
09:07:04 4 is the general concept.

09:07:07 5 Q. Did you prepare a slide that illustrates the time
09:07:10 6 value of the revenue Akoustis received 55 months early?

09:07:13 7 A. I did.

09:07:14 8 Q. Mr. Faison, can you please pull up PDX-11.4.

09:07:21 9 A. What you will see here is a bar graph, and what this
09:07:24 10 does is take Akoustis's annual revenues as it reported in
09:07:29 11 its SEC filings, this is the filing with the Securities and
09:07:34 12 Exchange Commission, on an annual basis starting in 2016,
09:07:37 13 run it through present. And it says if, in fact, these
09:07:43 14 revenues would have been delayed by 55 months, what would be
09:07:48 15 the differential in the time value of that money of having
09:07:53 16 it later, shifted out 55 months versus earlier.

09:08:00 17 Q. So is your damages calculation the money itself that
09:08:03 18 Akoustis received?

09:08:05 19 A. It's not. And this is important. It's not the
09:08:08 20 revenue. Under this analysis, Akoustis keeps the revenue
09:08:13 21 that it's earned. It's really only measuring the benefit --
09:08:16 22 the benefit by virtue of examining the time value, the
09:08:20 23 differential in having that revenue earlier. And I have
09:08:24 24 measured that using an interest calculation.

09:08:27 25 And I guess another thing to note, what this

Bennis - direct

09:08:30 1 analysis also assumes is that Qorvo -- or rather Akoustis
09:08:35 2 could have gotten in the market, it could have actually
09:08:39 3 developed a saleable product, it could have made all the
09:08:42 4 same sales, it could have accomplished the exact same growth
09:08:46 5 rate despite the fact that the revenues would be shifting
09:08:49 6 into a point in time when the start of the revenue
09:08:52 7 generation would have been right in the middle of the Covid
09:08:56 8 pandemic, so there are some general assumptions, if you
09:09:01 9 will, about shifting the money forward as is.

09:09:03 10 Q. Mr. Faison, can you please pull up PDX-11.5.

09:09:08 11 Ms. Bennis, does PDX-11.5 show the analysis that
09:09:12 12 you performed in this case?

09:09:13 13 A. It does. And I realize this looks very busy. And
09:09:17 14 I'm not going to walk through all of the numbers. But this
09:09:19 15 is the schedule that appeared within my report that really
09:09:22 16 kind of shows an interest calculation and how one works.

09:09:27 17 So I'll just point out a few important things
09:09:31 18 that will hopefully make a little bit more sense.

09:09:35 19 You'll see that these are again, these annual
09:09:40 20 revenue streams, row C, total Akoustis revenue. In 2016,
09:09:46 21 Akoustis reported \$254,844 in revenue. Those revenue
09:09:51 22 streams as reported run all the way through present. The
09:09:55 23 summation of all those equals 113.7 million in revenue that
09:10:00 24 Akoustis has reported.

09:10:03 25 Below that are all the different interest

Bennis - direct

09:10:07 1 calculations, and an interest rate you'll see of 14.8. 14.8
09:10:12 2 is important because that is the -- that was the active
09:10:16 3 interest rate that I employed as of present day in order to
09:10:20 4 bring all of those revenue streams up to the present value.
09:10:25 5 In order to evaluate as of a given point in time, what the
09:10:29 6 value of that revenue would be such that I can compare it to
09:10:32 7 what that revenue would be if the same revenue streams were
09:10:37 8 shifted forward 55 months.

09:10:39 9 Q. Why did you choose to use 14.8 percent as your
09:10:43 10 interest rate?

09:10:44 11 A. 14.8 percent is the -- it's the weighted average cost
09:10:50 12 of capital, commonly abbreviated was a WACC, rate of
09:10:56 13 Akoustis, so it comes directly from Akoustis's financial
09:11:00 14 model, it reflects their debt and equity structure. It's an
09:11:05 15 interest rate that companies use all the time because it
09:11:07 16 reflects their own unique financing abilities and position.
09:11:12 17 They use it to evaluate investment decisions and to
09:11:16 18 generally make money on how to finance their operation and
09:11:20 19 their decisions.

09:11:20 20 Q. And did you use Akoustis' WACC in the economic
09:11:25 21 formulas that are shown on the slide to determine the
09:11:28 22 present value of the revenues that Akoustis received, and
09:11:30 23 the benefit of the 55-month head-start?

09:11:34 24 A. I did. So again, on this particular slide, you'll
09:11:38 25 see that once the interest rate of 14.8 percent as of

Bennis - direct

09:11:42 1 present day, as of the time of this analysis, is applied to
09:11:46 2 the 113 million in total Akoustis revenues, the present
09:11:51 3 value of having all of those revenue streams as early as
09:11:55 4 2016, means that that same 113 million today is worth 140
09:12:01 5 million. And if you think about that, that makes sense
09:12:03 6 because they had the benefit of saying 2016 that \$254,000
09:12:09 7 number, if you look down at the bottom, in today's dollars
09:12:13 8 over all those additional years, is actually worth 821,508.
09:12:20 9 Because they have the money, and they have been able to
09:12:22 10 invest it and do things with it, that's the concept in
09:12:25 11 economics about the value of a dollar.

09:12:27 12 Q. Once you determine the present value of the revenue
09:12:30 13 that Akoustis actually received, what did you do next?

09:12:35 14 A. So next, I performed the same exercise, and in fact I
09:12:40 15 think I have another exhibit from my report. Yes.

09:12:45 16 So this took all of those annual revenue streams
09:12:49 17 and simply shifted them forward 55 months. So you'll know
09:12:53 18 that because, say in the first column you'll see the date is
09:13:00 19 2022, it used to be 2016, we shifted those revenue streams
09:13:05 20 55 months into the future. You know the revenue stream is
09:13:08 21 the same because we're still looking at the same
09:13:12 22 \$113 million figure, but instead of that \$254,834 number
09:13:18 23 that you see in the first column, being 800-some thousand
09:13:23 24 like we saw in the last slide, because it's now a lot closer
09:13:26 25 to present, that same revenue stream is only worth \$436,000,

Bennis - direct

09:13:33 1 highlighted down at the bottom. That's the difference of
09:13:35 2 having that money for a lesser period of time as we get
09:13:39 3 closer to present.

09:13:40 4 And then similarly, because we're measuring the
09:13:44 5 money as of present value, or as of now, because certain of
09:13:47 6 these revenue streams got pushed out into 2025, 2026, '27,
09:13:52 7 '28, we actually have to discount those back, because in the
09:13:56 8 real world, Akoustis wouldn't have had those yet. So those
09:14:00 9 get -- those get reduced by the same interest calculation
09:14:04 10 and therefore the present value of those revenue streams
09:14:08 11 equals 74 million, 74.8 million as you'll see. So that's
09:14:15 12 the time value of money concept.

09:14:16 13 Q. So how did you use the present value of the revenue
09:14:20 14 that Akoustis actually received, and the present value of
09:14:23 15 the revenue that Akoustis would have received without
09:14:27 16 misappropriating Qorvo's trade secrets in order to calculate
09:14:31 17 Qorvo's damages in this case?

09:14:33 18 A. So I think the next slide will show you, and it
09:14:36 19 really becomes quite simple.

09:14:38 20 You simply take the difference, look at the
09:14:41 21 differential between one revenue stream and the other, and
09:14:44 22 the difference represents again, not the revenue dollars
09:14:47 23 themselves, but just the benefit of having those at an
09:14:51 24 earlier point in time. So the benefit is measured of the
09:14:54 25 head-start to be the \$66,114,093 figure.

Bennis - direct

09:15:01 1 Q. Ms. Bennis, the slide at PDX-11.7 says date of
09:15:06 2 original analysis, 11/23. Can you tell the jury what that
09:15:09 3 means?

09:15:09 4 A. Yeah. So I mentioned that my report analysis was
09:15:12 5 finalized and issued back in November. And so at that point
09:15:16 6 in time, we knew that the trial, the decision of the jury
09:15:23 7 would be happening now, so it made sense to run the analysis
09:15:26 8 through now. But at that point in time in November, fiscal
09:15:30 9 year 2024, which we heard testimony last week runs from
09:15:36 10 July 1st through June 30th of a year, was not yet complete.
09:15:40 11 So in fact, if you go back a couple of slides, you'll see --

09:15:45 12 Q. 11.5.

09:15:47 13 A. That last column 2024, EST, EST stands for estimated.
09:15:53 14 The estimate comes from Akoustis's own files, this is what
09:15:56 15 Akoustis projected that it was going to earn in it's fiscal
09:15:59 16 year 2024 on a whole.

09:16:06 17 Q. Have you performed any updated calculations based
09:16:10 18 upon information presented to you during trial?

09:16:13 19 A. Yes. So last week when I was listening like you, I
09:16:18 20 heard the testimony of Mr. Aichele that suggested that their
09:16:22 21 -- that Akoustis's fiscal year 2024 results were going to
09:16:27 22 be 28 million in total, which was different than the
09:16:30 23 projections that they had made back that were available at
09:16:34 24 the time of my report. So for the benefit of the jury to
09:16:37 25 see what that would look like, I ran that number through the

Bennis - direct

09:16:41 1 model. When you run that projection, it's not complete, so
09:16:45 2 we don't know if that's ultimately how it will end up, but
09:16:48 3 if you run that figure through the model, performing the
09:16:53 4 same economic analysis, you get a total head-start damages
09:16:59 5 figure of 50.3 million.

09:17:03 6 Q. Ms. Bennis, I want to make sure that your testimony
09:17:06 7 for the jury is clear. The calculation that you provided to
09:17:08 8 PDX-11.7, which is the prior slide, Mr. Faison, was based
09:17:13 9 upon information that you received from Akoustis's own files
09:17:17 10 relating to what it thought it's revenue streams were going
09:17:21 11 to be for 2024; is that correct?

09:17:24 12 A. That's correct.

09:17:24 13 Q. And if we flip to the next slide, 11.8, you have
09:17:28 14 updated those calculations within your own model based upon
09:17:32 15 testimony that Mr. Aichele provided to the jury, is that
09:17:35 16 correct?

09:17:35 17 A. That's correct.

09:17:38 18 Q. When you calculated the unjust enrichment benefit to
09:17:43 19 Akoustis, did you consider how the trade secrets impacted
09:17:46 20 Akoustis's costs?

09:17:47 21 A. Yes. So recall when Dr. Shanfield talked about those
09:17:51 22 analysis, that he really only talked about the benefit of
09:17:54 23 the head-start in -- by virtue of the time saved. But in
09:17:58 24 reality, companies incur real research and development hard
09:18:04 25 expenses. So this analysis again, is only looking at the

Bennis - direct

09:40:40 1 that the trade secret misappropriation, of the confidential
09:40:44 2 information is really -- it's affected the business
09:40:49 3 strategies, the design of the filters, the testing, all of
09:40:53 4 the different elements that help actually build and test and
09:40:56 5 make a product successful.

09:40:58 6 Q. Mr. Faison, could you please go back to PDX-11.2.

09:41:02 7 MS. AYERS: Your Honor, may I approach?

09:41:05 8 THE COURT: You may.

09:41:17 9 BY MS. AYERS:

09:41:25 10 Q. Ms. Bennis, do you recognize the document that I just
09:41:27 11 handed you?

09:41:28 12 A. I do.

09:41:28 13 Q. Can you explain to the jury what it is?

09:41:31 14 A. Yes. So this is Appendix C, as it was included in my
09:41:36 15 report. And what I did here was, with respect to the trade
09:41:40 16 secret misappropriation and unfair competition -- or
09:41:43 17 misappropriation of confidential information analysis, is
09:41:47 18 because Dr. Shanfield's analysis considered a period of
09:41:52 19 55 months, but he also included really the benefit in
09:41:57 20 monthly increments. I have actually run these monthly
09:42:01 21 increments through the model such that there is an analysis
09:42:05 22 for damages under a head-start period of one month all the
09:42:09 23 way up to 54 months, in addition to the 55 months. So this
09:42:13 24 is the appendix that simply runs the figures through the
09:42:18 25 model in monthly increments.

Bennis - cross

09:43:38 1 just make sure that the jury understands the construct for
09:43:43 2 damages experts testifying in court. And you would agree
09:43:49 3 with me that the construct in any lawsuit, let's say it's
09:43:54 4 intellectual property, products liability, employment, is
09:43:57 5 that there are no -- there is no damages without liability;
09:44:02 6 correct?

09:44:03 7 A. Yes.

09:44:04 8 Q. But because the jury typically has not heard all of
09:44:09 9 the evidence from both sides, especially when the Plaintiff
09:44:12 10 puts on its case, the damages experts are asked to assume
09:44:18 11 liability for purposes of their testimony so that if the
09:44:24 12 jury having heard all of the evidence finds liability, then
09:44:28 13 they have an understanding about how to assess damages, if
09:44:31 14 any; is that correct?

09:44:32 15 A. Yes.

09:44:34 16 Q. So both you and Ms. Irwin, Carolyn Irwin, Akoustis's
09:44:40 17 damages expert, come in assuming liability, but you're not
09:44:44 18 testifying as to liability, correct?

09:44:46 19 A. Correct. I did mention that at the beginning, yes.

09:44:48 20 Q. So let me just recap. In essence, you have three
09:44:55 21 different damages opinions. You have one as to trade secret
09:44:58 22 misappropriation; correct?

09:45:00 23 A. Yes.

09:45:00 24 Q. And that is based on an unjust enrichment theory
09:45:06 25 using a head-start. And that is for \$66.1 million; correct?

Bennis - cross

09:45:11 1 A. Yes.

09:45:13 2 Q. And then you have an opinion regarding unfair
09:45:18 3 competition damages. The first part of that is the benefit
09:45:22 4 of the head start to Akoustis, but that's from both trade
09:45:29 5 secrets and confidential information, correct?

09:45:34 6 A. Yes. Trade secrets as I understand it are subsumed
09:45:37 7 in confidential information, yes.

09:45:39 8 Q. Okay. So there is a bucket there of confidential
09:45:44 9 information, some of which are trade secrets and some of
09:45:47 10 which is just confidential information, but not trade
09:45:50 11 secrets?

09:45:50 12 A. Yes, generally.

09:45:51 13 Q. And that, your opinion for that also is based on an
09:45:56 14 unjust enrichment theory based on a head start, and it's
09:46:00 15 \$66.1 million?

09:46:01 16 A. That's right.

09:46:02 17 Q. And there is also the second component of unfair
09:46:06 18 competition is the poaching damages, those are direct
09:46:10 19 damages to Qorvo; correct?

09:46:11 20 A. Correct.

09:46:12 21 Q. And that's 809,000?

09:46:14 22 A. Yes.

09:46:15 23 Q. And then number 3, you have the patent infringement
09:46:18 24 damages of \$279,000?

09:46:20 25 A. Correct. Sounds right.

Bennis - cross

09:49:30 1 A. That's my understanding.

09:49:32 2 Q. Okay. You have calculated no damages for Qorvo's
09:49:36 3 actual loss, correct?

09:49:36 4 A. Correct, I evaluated unjust enrichment, right.

09:49:40 5 Q. You didn't attempt to calculate or to measure the
09:49:44 6 damages caused by misappropriation alleged against Akoustis
09:49:50 7 as measured by imposition of a reasonable royalty either,
09:49:55 8 correct?

09:49:55 9 A. I did not reevaluate the reasonable royalty for the
09:49:59 10 claims of trade secret, that's correct.

09:50:00 11 Q. And you don't really have a specific reason why you
09:50:02 12 did not express an opinion regarding the amount of Qorvo's
09:50:09 13 actual damages, correct?

09:50:11 14 A. I think when we're phased with the task of evaluating
09:50:17 15 damages, sometimes there are choices to be made about what's
09:50:20 16 the most saleable, there is no question, there was no
09:50:27 17 question once the evidence kind of was laid out, the benefit
09:50:31 18 to Akoustis of obtaining this information. And so because
09:50:36 19 there was such a clear head start benefit, it made sense to
09:50:40 20 evaluate the unjust enrichment accordingly.

09:50:43 21 There is no question that there are actual
09:50:46 22 losses to Qorvo. The loss of the confidential nature of its
09:50:52 23 own information. But for purposes of the damages analysis
09:50:55 24 that I performed, I focused on that second prong, and that
09:50:58 25 is unjust enrichment enjoyed by Akoustis.

Bennis - cross

09:51:01 1 Q. But since they're additive, Qorvo could have claimed
09:51:05 2 more damages if it had actual damages to show, you just
09:51:09 3 didn't do that, correct?

09:51:10 4 A. So, I made several conservatives in my analysis.
09:51:15 5 Like I mentioned, there were elements of unjust enrichment
09:51:19 6 that I also did not include.

09:51:20 7 Q. And unjust enrichment measures the benefit realized
09:51:25 8 by the Defendant as a result of its use of the trade
09:51:28 9 secrets, correct?

09:51:29 10 A. Yes.

09:51:29 11 Q. There is no benefit without use, right?

09:51:33 12 A. The unjust enrichment in this case follows the --
09:51:39 13 yes, the benefit from the use, yes, agree.

09:51:47 14 Q. So let's focus now on unjust enrichment, which is
09:51:51 15 what you did calculate with respect to the trade secret
09:51:55 16 misappropriation element here.

09:51:56 17 And you would agree with me that court cases and
09:52:01 18 literature in the area recognizes three primary categories
09:52:07 19 of unjust enrichment in trade secret cases, correct?
09:52:13 20 Defendants' profits is one. Head start benefit is another,
09:52:19 21 and avoided costs is the third. Right?

09:52:27 22 A. I certainly have done lots of reading and analysis
09:52:30 23 that evaluate unjust enrichment in a variety of ways. Those
09:52:35 24 are certainly some of the ways that I have seen, yes, in
09:52:41 25 thinking about it.

Bennis - cross

09:54:05 1 said in the past it has -- it has showed a small gross
09:54:11 2 profit; correct?

09:54:13 3 A. I am aware that Akoustis has shown a small gross
09:54:17 4 profit.

09:54:18 5 Q. We'll look at a table from your report so that we can
09:54:22 6 make that clear.

09:54:22 7 So you understand what avoided costs are?

09:54:28 8 A. Yes.

09:54:28 9 Q. And it's the costs -- they would be the development
09:54:34 10 costs that a Defendant avoids by using plaintiffs' trade
09:54:41 11 secrets; correct?

09:54:43 12 A. Yes, generally speaking.

09:54:46 13 Q. And I assume that you understand that avoided costs
09:54:52 14 is also a way to measure the benefit to a Defendant from a
09:54:57 15 head start?

09:54:59 16 A. Yes, I think I mentioned that.

09:55:01 17 Q. You did not attempt to calculate the benefit to
09:55:06 18 Akoustis from an-- on the basis of an avoided cost
09:55:11 19 methodology, correct?

09:55:12 20 A. So like I mentioned earlier, my calculations
09:55:17 21 evaluated the time element of that benefit, it did not
09:55:21 22 evaluate any of the hard research development expenses that
09:55:25 23 are typically incurred by businesses.

09:55:27 24 Q. So that would be yes, you did not calculate an
09:55:30 25 avoided cost calculation?

Bennis - cross

09:55:31 1 A. I did not add into my damages overall calculations,
09:55:35 2 those types of expenses.

09:55:36 3 Q. And we spoke at length in January during your
09:55:40 4 deposition, do you remember that?

09:55:41 5 A. I do.

09:55:42 6 Q. And you suggested the reason for not attempting to
09:55:46 7 calculate avoided costs was that you did not have the
09:55:49 8 information to calculate that, do you recall that?

09:55:52 9 A. So in instances like this one, and in general,
09:55:58 10 Akoustis -- there are no such records in the files of
09:56:02 11 Akoustis because it didn't do it. It didn't engage in the
09:56:04 12 research and development expenses that it instead
09:56:08 13 misappropriated.

09:56:09 14 So there are instances where one can use, for
09:56:13 15 instance, like in this case, Qorvo's research and
09:56:17 16 development expenses as a proxy or as an estimate for what
09:56:22 17 it would take for Akoustis or a company in its position to
09:56:26 18 do the same thing. They're Qorvo's trade secrets, so I know
09:56:29 19 we heard testimony last week that Qorvo spent a billion
09:56:36 20 dollars in BAW filter research and development over the
09:56:40 21 course of twenty years. Just to finish the analysis --

09:56:42 22 Q. Well, you're not answering my question. In fairness,
09:56:47 23 I asked you, didn't you say that, and you went off on a
09:56:50 24 tangent to put in front of the jury information that you
09:56:52 25 wanted it to hear. So you know, your counsel can always

Bennis - cross

10:05:53 1 capital equipment. It's being consumed by the costs of
10:05:57 2 revenue. And that's really the problem with your theory is
10:06:00 3 you're trying to infer a benefit from revenue without taking
10:06:06 4 real world facts into account; right?

10:06:09 5 A. No, I disagree completely.

10:06:12 6 Q. Okay.

10:06:13 7 A. And again --

10:06:14 8 Q. Let's say you're aware that the courts rely on
10:06:25 9 publications that have considered expert witness
10:06:29 10 methodologies for estimating a head start benefit to a
10:06:33 11 Defendant having endorsed expert opinions using the amount
10:06:36 12 of the Defendant's incremental products, that's the
10:06:42 13 Defendant's profits, right?

10:06:43 14 A. I believe so, yes.

10:06:45 15 Q. And in a head start, they have also approved the use
10:06:50 16 of the increased value of the Defendant's company, company
10:06:56 17 value, as a result of a head start it obtained, you're aware
10:07:00 18 of that, right?

10:07:01 19 A. Which is what all these types of things get to, yeah.

10:07:05 20 Q. But you did not perform a valuation of Akoustis as a
10:07:11 21 company, correct?

10:07:12 22 A. My analysis focuses just on the isolated benefit.

10:07:16 23 Q. Right. And you're aware, because I asked you at your
10:07:21 24 deposition, you're aware of the Sedona Conference, right?

10:07:25 25 A. Yes.

Bennis - cross

10:11:33 1 MR. ELKINS: Your Honor, we would move the
10:11:35 2 Sedona Conference Journal Volume XIX, Number 2, into
10:11:39 3 evidence.

10:11:41 4 MS. AYERS: Objection, Your Honor.

10:11:41 5 THE COURT: Objection sustained.

10:11:45 6 BY MR. ELKINS:

10:11:45 7 Q. Let's look --

10:11:46 8 THE COURT: It's inconsistent with the portion
10:11:53 9 of the 803 series of rules.

10:11:57 10 BY MR. ELKINS:

10:11:58 11 Q. Can you go to page 39, 39 of 81, which is not here,
10:12:04 12 but I think it's at page 684. Well, let me do this without
10:12:36 13 the document.

10:12:38 14 When we spoke at your deposition in January, you
10:12:48 15 could not recall at that time having ever previously
10:12:51 16 performed a head start estimate of benefit in a trade secret
10:13:01 17 case before this one, is that accurate?

10:13:06 18 A. A head start benefit?

10:13:11 19 Q. Yes.

10:13:17 20 A. I'm not sure that I have.

10:13:21 21 Q. And while you're -- you're looking at the -- at least
10:13:28 22 in direct examination, you said that you're looking at the
10:13:31 23 time value of money received by Akoustis. The money
10:13:39 24 received is revenue in this case; correct?

10:13:42 25 A. Yes.

Bennis - cross

10:13:42 1 Q. And you could not at your deposition, identify a
10:13:49 2 single court case where a court approved using a Delta N
10:13:56 3 revenue as adjusted for the time value of money as a basis
10:14:01 4 for estimating a head start benefit, correct?

10:14:05 5 A. That's right.

10:14:06 6 Q. And you cannot identify a single informative
10:14:11 7 publication approving using revenue as the basis for
10:14:16 8 estimating a head start benefit in a trade secret case;
10:14:21 9 correct?

10:14:24 10 A. I mean, I certainly can't cite to every publication
10:14:28 11 that's ever been authored. I can't think of any examples.
10:14:31 12 I think every analysis that I see, both that I perform and
10:14:35 13 others in my shoes perform, are unique to the facts and
10:14:38 14 circumstances present in the analysis that we're asked to
10:14:41 15 perform. So what I did in this particular case, was what I
10:14:47 16 felt fit best with the facts.

10:14:51 17 Q. Even though there was no prior basis for doing it;
10:14:55 18 right?

10:14:55 19 A. Well, like we have discussed, I mean, the measure of
10:14:58 20 a head start benefit is right in line with what's
10:15:04 21 appropriate to evaluate in the case of trade secret
10:15:07 22 misappropriation like we have here.

10:15:10 23 Q. Now, in your but for world, in your but for world is
10:15:18 24 what damages experts call what should have happened but for
10:15:24 25 the facts that give rise to liability, in this case trade

11:10:53 1 of the research and development expenses that Akoustis
11:10:56 2 incurred. What was the majority of those research and
11:11:01 3 development expenses incurred?

11:11:04 4 A. According to this particular schedule, you can see
11:11:07 5 that those, the majority that has been incurred in preceding
11:11:14 6 years.

11:11:15 7 Q. For the benefit of the jury, can you point out what
11:11:18 8 Akoustis research and development expenses were in 2016?

11:11:22 9 A. \$1,758,701.

11:11:28 10 Q. What are the research and development expenses that
11:11:30 11 Akoustis is incurring right now in 2023 and 2022?

11:11:33 12 A. Fiscal year 2022, it recorded approximately
11:11:38 13 35.7 million. Fiscal year 2023, it recorded approximately
11:11:44 14 33.2 million.

11:11:45 15 Q. Mr. Elkins also asked you questions about the 104
11:11:50 16 trade secrets that Qorvo initially asserted against Akoustis
11:11:56 17 which have now for efficiency sake for the jury been dropped
11:12:01 18 down to somewhere in the range of the forties. Was your
11:12:04 19 original head start analysis that you performed in November
11:12:08 20 based upon Groups 1 through 8 of Dr. Shanfield's report?

11:12:13 21 A. That's correct.

11:12:14 22 Q. And was your analysis based upon the 55-month head
11:12:18 23 start that Dr. Shanfield opined to during trial?

11:12:22 24 A. Yes.

11:12:24 25 MS. AYERS: I have no further questions for this

1 have been adjusted and some steps are being taken out of the
2 instructions right now. So I'll start with lead counsel for
3 the defense. Hopefully I have got the right lead counsel.

4 MR. ELKINS: We will --

5 MR. LEMIEUX: I'll defer to my colleague.

6 THE COURT: I think that's what he told me, is
7 that right? We're set to go.

8 MR. ELKINS: Your Honor, we have two brief
9 Rule 50(a) motions to bring. The first one we could not
10 bring until we heard testimony, and I will be extremely
11 brief.

12 The basis of this motion, Your Honor, is that
13 the head start duration estimated by Dr. Shanfield coupled
14 with the head start damages opinion from Ms. Bennis relying
15 on Dr. Shanfield's defective, in our view, estimate may not
16 be considered by the jury. Together, they are the bases of
17 Qorvo's head start benefit theory. And together, they do
18 not meet the standard for the admission of expert evidence.

19 I am not going to repeat the admissibility
20 tenants of Rule 702. The Court is well aware of them. But
21 I do believe that one, in this Court earlier this year, the
22 Court stated to satisfy the reliability requirement, the
23 expert testimony must be based on the methods and procedures
24 of science, not on subjective belief and unsupported
25 speculation. And that is of extreme relevance in this case.

1 Ms. Bennis's calculation is entirely premised on
2 Dr. Shanfield's conclusion that a 55-month head start is
3 reasonable. Dr. Shanfield admitted on cross-examination at
4 trial that he has no facts or publications to support his
5 55-month head start calculation, or the individual trade
6 secret by month calculations that comprise it.

7 While he considered where he could find
8 information that would give him a reference point for each
9 trade secret period, he admitted that he relied on his own
10 belief about things. And I am citing to the trial record at
11 1381 through 1403.

12 Those beliefs are an improper basis for expert
13 opinion under Federal Rule of Evidence 702.

14 He relied on his own work at Draper Labs, which
15 he admitted was not a manufacturing company; his
16 conversations with unnamed people at Qorvo, although he did
17 identify Dr. Aigner, though he admitted Dr. Aigner did not
18 know for some categories. He relied on his own corporate
19 experience which ended in 2003, more than twenty years ago.
20 He relied on an unnamed publication that was not in his
21 report. And he relied on an old colleague with whom he
22 spoke, also not in his report.

23 Nothing about his testimony -- nothing about his
24 head start duration opinion as provided in his opinion is
25 reliable, and therefore it's inadmissible.

1 And for the reasons set forth in our prior
2 Daubert motion, Ms. Bennis's head start benefit opinion is
3 based on revenue rather than profit. It does not meet the
4 standards for the admission of opinion testimony under 702,
5 but is rendered even more unreliable by her reliance on
6 Dr. Shanfield's head start duration estimate.

7 And that, too, expert witnesses are combining
8 unreliable opinions to spin a tail of head start fantasy
9 renders the Court's gatekeeping role even more imperative in
10 this case than it usually is. The Court should instruct the
11 jury to disregard the two opinions, and without admissible
12 evidence regarding the value of Akoustis's purported head
13 start benefit, that issue should be removed from the jury's
14 purview.

15 That's all I have on that, Your Honor.
16 Mr. Lemieux has a related, but separate motion.

17 THE COURT: Okay. We will -- before we do that,
18 we'll hear a brief response on this head start question.

19 MS. AYERS: Thank you, Your Honor.

20 As you know, Rule 703 allows an expert's
21 opinions to be based on fact and data that the expert has
22 personally observed in this case. Dr. Shanfield testified
23 at length regarding the misappropriation. He also testified
24 at length regarding how he calculated his head start
25 55-month analysis. Through his testimony, he identified

1 formulas that accountants use every day in order to actually
2 analyze and measure the time value of revenue that a company
3 receives.

4 THE COURT: All right. This has been
5 established and addressed to some degree already, and so
6 time value analysis is very important in a case. And the
7 unjust enrichment in light of the standard will be denied.
8 We're ready for the second matter.

9 I will say this, this is something we really
10 worked on. I wish I had a case that told us exactly what to
11 do. All of us have looked for that, and we have not been
12 able to do that, so it's an interesting and challenging
13 question. It is a challenging question. Counsel, you got
14 the second part.

15 MR. LEMIEUX: It's a related 50(a) motion, Your
16 Honor. That basically under the North Carolina Unfair
17 Competition Act that has been alleged in this case, it has
18 three requirements. One, that there be an unfair deceptive
19 act or practice. Two, that it will be in or affecting
20 commerce. And three, that it proximately causes actual
21 injury to the claimant. As we heard Ms. Bennis' testimony,
22 she did not provide any opinion whatsoever on actual injury
23 to the Plaintiff in this case.

24 They chose to follow an unjust enrichment which
25 is just simply the benefit allegedly received by the

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09:01:13 1 **BY MR. LEMIEUX::**

09:01:13 2 **Q.** Has Dr. Heinrich, then, Professor Nguyen, introduced
09:01:16 3 a new variable that has to be accounted for during this
09:01:20 4 cleaving process?

09:01:21 5 **A.** Yes. I would say yes.

09:01:24 6 **Q.** If we can now look back at exhibit --
09:01:26 7 Demonstrative 2.03.

09:01:28 8 Now, you've mentioned a second requirement for
09:01:31 9 Element 1C, Professor Nguyen. It is also not infringed,
09:01:35 10 in your opinion, by the Akoustis devices.

09:01:37 11 Could you explain what you mean by that?

09:01:39 12 **A.** Yes. I think we have a demonstrative to help with
09:01:42 13 us.

09:01:42 14 **MR. LEMIEUX:** Can we have Demonstrative 2.05,
09:01:45 15 please.

09:01:45 16 **THE WITNESS:** Yes. So this is the simulation
09:01:47 17 that was submitted to claim that this last part of that
09:01:55 18 Element 1C in the '018 patent was infringed. However, if
09:02:01 19 you look at this data, there are some significant problems
09:02:04 20 with that data.

09:02:06 21 So first of all, I can point out a couple of
09:02:08 22 things, right. There's no units on this. I think that's
09:02:10 23 been pointed out before. That's a pretty big deal,
09:02:12 24 actually. Let me say more about this later on when we
09:02:14 25 talk about the other patent.

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09:02:15 1 For now, if you look at this data, what they
09:02:18 2 did was they sort of looked at the right side of this, saw
09:02:20 3 that the green was larger than the blue. And I think
09:02:23 4 there is an animation we can put in.

09:02:24 5 Yeah. So they looked at that part, saying the
09:02:26 6 green was larger than the blue. But you've got to look at
09:02:31 7 this part, too. You can't just look at one event. So if
09:02:33 8 we circle this part right here, the blue is higher than
09:02:36 9 the green. So you can't even make much out of this in
09:02:38 10 terms of claiming that green is necessarily higher than
09:02:41 11 the blue.

09:02:41 12 But, you know, all of that is nonsense. I
09:02:43 13 don't have to talk about any of this. It would be a
09:02:45 14 complete waste of time, because look at the Y axis of
09:02:48 15 this. It is lateral energy flux, plotting lateral energy
09:02:52 16 flux versus frequency. Remember what the patent requires
09:02:55 17 is that the bandwidth increase. This is not bandwidth.
09:02:59 18 They're not measuring bandwidth. They needed to measure
09:03:04 19 bandwidth. And so what do I mean by "bandwidth." I have
09:03:08 20 a demonstrative that can help explain.

09:03:10 21 So this is a plot that I think you've seen
09:03:13 22 several times over this trial. And so what this plot is,
09:03:15 23 is sort of the energy through the device as a function of
09:03:19 24 frequency.

09:03:21 25 And so if this is the filter -- so each device

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09:05:45 1 **A.** Oh, yeah.

09:05:45 2 **Q.** And in this case, Dr. Shanfield stated that he did
09:05:50 3 not do any actual testing or measurements, and I believe
09:05:53 4 Dr. Bravman agreed with that.

09:05:55 5 Instead, what did they do?

09:05:57 6 **A.** They, instead, did simulations like the one that you
09:06:01 7 showed there. And that simulation doesn't even have much
09:06:03 8 to do with bandwidth. It has more to do with loss, which
09:06:07 9 are different quantities of this plot here. And it's even
09:06:09 10 difficult to determine how much loss you've got out of the
09:06:13 11 leakage of it. You certainly cannot determine the
09:06:14 12 bandwidth off of that.

09:06:19 13 **Q.** So instead of the proxy simulations that
09:06:24 14 Dr. Shanfield performed, he should have been able to
09:06:27 15 physically measure the bandwidth.

09:06:31 16 Could we go back to -- well, first, let me ask you,
09:06:35 17 Professor Nguyen: Are you aware of any measurements that
09:06:38 18 were done by the plaintiff in this case related to the
09:06:41 19 bandwidth that were performed on the actual accused
09:06:45 20 devices?

09:06:45 21 **A.** If they measured it, we were not supplied with the
09:06:50 22 measurements.

09:06:50 23 **Q.** Now, are you aware of whether Qorvo or its experts
09:06:54 24 had samples of the accused devices on which they could
09:06:57 25 have performed actual physical bandwidth testing?

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09:19:45 1 do with loss. So I can't say it's completely
09:19:48 2 inappropriate. But let's take a look at what we're
09:19:51 3 looking at here. So this is lateral energy flux versus
09:19:54 4 frequency.

09:19:55 5 These lines, vertical lines indicate the
09:19:58 6 passband for the resonator itself. And the blue curve
09:20:02 7 corresponds to the offending device, the accused device,
09:20:06 8 which means the device for which N is not equal to 1. The
09:20:10 9 green curve corresponds, I guess, a reference device where
09:20:14 10 N is equal to 1. So this device in green would not
09:20:20 11 infringe the patent.

09:20:22 12 On the first set of data that was sent, notice
09:20:25 13 how the blue curve is much larger than the green curve?
09:20:28 14 What this means is the blue curve, the device that
09:20:31 15 actually satisfies and infringes the patent -- well, not
09:20:35 16 infringes the patent, but satisfies the element where N is
09:20:38 17 not equal to 1, actually is losing more energy than the
09:20:42 18 green one.

09:20:43 19 So in this actual curve, this is saying that
09:20:47 20 this device does not infringe the patent. It's outright
09:20:50 21 saying that. And I put that in my initial report.
09:20:52 22 Interestingly enough, a month later -- I'm not sure how
09:20:57 23 many weeks later after this, this data came in. And you
09:21:02 24 heard that it was blamed on a clerical error of some sort.
09:21:07 25 Right?

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09:21:08 1 And so in this data now, if you look at some
09:21:09 2 parts of this, oh, there's a part where the green is
09:21:12 3 greater than the blue. This is the part that the Qorvo
09:21:16 4 experts pointed out as, you know, the reference device
09:21:21 5 being worse than the accused device because it has more
09:21:25 6 energy. But, you know, you have to look at all of this
09:21:27 7 range. You can't just look at this portion. You've got
09:21:29 8 to look at this portion, you've got to look at that
09:21:32 9 portion.

09:21:33 10 So if we can move the animation forward.

09:21:35 11 What you really have to do is look at the
09:21:37 12 shaded regions here. Which has more of the shaded
09:21:40 13 regions? So on this one, obviously there's more of this
09:21:42 14 shaded region where the blue is higher than the green.
09:21:46 15 Obvious, here, this device does not infringe, according to
09:21:49 16 this simulation.

09:21:50 17 If you go back to this here, I don't know. Is
09:21:53 18 the blue more than the green? Is the green more than the
09:21:56 19 blue? I'm not sure I can tell. And, you know, even if
09:21:58 20 you wanted to count all the boxes, you know, sum up the
09:22:03 21 boxes, say which one has more boxes than the other -- so
09:22:05 22 the little boxes, I'm talking about the grid there.

09:22:08 23 You still couldn't conclude that here because
09:22:10 24 let's look at the rest of this plot. There are no units
09:22:14 25 here. That is a huge issue. So think of yourself, say,

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09:22:20 1 wanting to bake a cake or something and you're asking your
09:22:23 2 friend, how much sugar do I need. And your friend says,
09:22:27 3 well, you need two.

09:22:28 4 The first thing that's going to come to your
09:22:30 5 mind is two what? Two tablespoons? Two teaspoons? Two
09:22:34 6 cups? How much sugar do I need? And without that, you're
09:22:38 7 not really going to be able to bake the cake. I guess you
09:22:41 8 could go forward. It may not taste fantastic.

09:22:45 9 And so it's no different than science and
09:22:47 10 engineering. You can't just put up a plot with no units
09:22:50 11 and say, all right, there it is. We need units to know
09:22:53 12 exactly what we're looking at. And, in fact, if you look
09:22:56 13 at units that you think it ought to be -- so the units it
09:22:59 14 ought to be for lateral energy flux is watts per square
09:23:02 15 meter. That is the international standard unit that you
09:23:05 16 would use.

09:23:06 17 And if I plug in watts per square meter here,
09:23:09 18 and I actually calculate out the watts, which is what I
09:23:11 19 really need to compare the power through this device, the
09:23:14 20 amount of watts this corresponds to because the device is
09:23:18 21 so small, is on the order of 10 to the minus 19. What
09:23:23 22 does 10 to the minus 19 mean?

09:23:25 23 You've already seen this 10 to the minus 10.
09:23:26 24 That's a very small number. What that means when you say
09:23:28 25 10 to the minus 10 is you're going to put nine zeros in

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front of the number. Say if I'm looking at this peak here, that looks like 3. I'm going to put a decimal point and nine zeros in front of that.

So in the power that I calculated, 10 to the minus 19, that would be three times 10 to the minus 19. Let me go ahead and count that out to you. That's 0.000000000000000000. That's a lot of zeros. This is a tiny signal that we're looking at.

And so this is being done by finite element simulation. Finite element simulation, I think you've heard from the Qorvo experts, is not an easy thing to do. This is something where you can try the simulation first, but you can't just run one simulation. Remember, you have to run several of them in order to get it right. That's because just small differences in the inputs you put into this -- in this case, it takes meshing. A small difference in the meshing, small difference maybe in -- anything you can put into this will change the result.

You've got to get it to the point that once you start making differences, changes, the result doesn't change much.

Well, the amount of excursion and signal here is so small that one needs to suspect whether the precision of the software is good enough to even delineate this with this. And when I look at these two curves,

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09:24:57 1 because this is such a small scale right here, I wonder if
09:25:01 2 this really is a correct simulation, not really a clerical
09:25:05 3 error, but just has some change in the meshing of the
09:25:07 4 finite element that yields this.

09:25:09 5 So same device, same two plots there. I wonder
09:25:13 6 about that because the precision of the software, I don't
09:25:17 7 think, is good enough to predict losses this small.

09:25:20 8 **BY MR. LEMIEUX:**

09:25:22 9 **Q.** So, Professor Nguyen, in your opinion, then, is the
09:25:26 10 testing that was performed by Dr. Shanfield sufficient to
09:25:30 11 prove the final element of the Claim 9 of the '755 patent?

09:25:36 12 **A.** It is not sufficient.

09:25:37 13 **Q.** And could you briefly summarize why the testing that
09:25:40 14 was performed by -- or the simulations, excuse me --
09:25:43 15 performed by Dr. Shanfield insufficient to actually prove
09:25:49 16 the power loss that it is allegedly measuring?

09:25:52 17 **A.** Okay. So three different things. One, no units on
09:25:55 18 these plots. Second, just the analysis here of trying to
09:25:59 19 get all the areas in these shaded regions I don't think
09:26:04 20 was done correctly. Finally, the total signal you're
09:26:07 21 looking at here is so small, I don't think the software
09:26:10 22 has the precision to predict this accurately.

09:26:14 23 **Q.** Now, Dr. Bravman, when he testified, he admitted that
09:26:15 24 he didn't know what the units were either on this vertical
09:26:18 25 index, and yet still offered an opinion of infringement.

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10:25:24 1 when you have accelerating losses. As you will see, it
10:25:28 2 makes no sense.

10:25:30 3 **Q.** What is your second opinion?

10:25:32 4 **A.** That a different damages approach called the avoided
10:25:37 5 costs method, which I will explain, is a better fit for
10:25:41 6 calculating any benefit that Akoustis gained from
10:25:46 7 misappropriating and using the trade secrets.

10:25:54 8 **Q.** Did you calculate Akoustis' avoided cost benefit?

10:25:56 9 **A.** I did. And, again, I think it was a better fit for
10:25:57 10 this case, and it ends up being around \$305,000.

10:26:05 11 **Q.** Do you have any opinions about Ms. Bennis' unfair
10:26:08 12 competition opinions in this case?

10:26:09 13 **A.** I do.

10:26:12 14 **Q.** What are they?

10:26:13 15 **A.** So her calculation of the employee poaching is
10:26:19 16 fundamentally flawed. First, she did no research on
10:26:22 17 whether or not Akoustis -- excuse me -- Qorvo actually
10:26:23 18 replaced the employees that were allegedly poached, yet
10:26:28 19 her calculation, at least she represented it to be the
10:26:31 20 cost that Qorvo incurred to replace those employees or
10:26:34 21 to -- the cost associated with losing them.

10:26:37 22 Second, she assumed that every one of the 19 people
10:26:40 23 she discussed with the HR director at Qorvo was poached.
10:26:45 24 Whereas, I understand that the plaintiff's argument here
10:26:49 25 is that employees were poached because they had access to

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10:32:44 1 it allegedly suffered?

10:32:46 2 **A.** No. I've not seen any evidence, and nobody has
10:32:49 3 pointed me to any lost sales by Qorvo.

10:32:53 4 **Q.** So did you see any issues with Ms. Bennis'
10:32:57 5 calculation of the head start benefit in this case?

10:33:01 6 **A.** Yes. There are three key issues -- I think if you go
10:33:04 7 to the next slide, they are summarized high level.

10:33:07 8 First, as we'll get into, she only focuses on
10:33:10 9 revenue. And that's just fundamentally flawed. She
10:33:14 10 doesn't include any costs related to the production of the
10:33:16 11 BAW filters. And then setting aside the fact that she
10:33:20 12 completely ignores costs, she's overstated revenue in two
10:33:24 13 important ways.

10:33:25 14 First, she's including revenue that had nothing to do
10:33:28 15 with the BAW filters that embody the alleged trade
10:33:31 16 secrets. And then, second, she, initially at least,
10:33:32 17 overstated the projected 2024 fiscal year revenue.

10:33:40 18 **Q.** I understand you have a slide with respect to the
10:33:42 19 last two points?

10:33:43 20 **A.** I do. So just with respect to the overstating of
10:33:46 21 revenue, she presented two numbers in her testimony. She
10:33:50 22 had a \$66 million number and a \$50 million number. The
10:33:54 23 second and third point I make here represents 54 percent
10:33:59 24 of her \$50 million calculation.

10:34:02 25 If you were to benchmark that off of her \$66 million

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10:34:05 1 calculation, this error -- these combined errors are
10:34:10 2 65 percent of her \$66 million, which is -- just
10:34:14 3 demonstrates the sensitivity of these two errors. Again,
10:34:18 4 setting aside the fact that she didn't look at costs at
10:34:21 5 all.

10:34:21 6 **Q.** Now, can we go back to the first point, ignoring
10:34:30 7 costs relating to production of BAW filters. Can you
10:34:32 8 explain what you mean by that in little bit more detail?

10:34:37 9 **A.** Yes. So it's important to understand that you can't
10:34:41 10 sell a product without making it. And she doesn't account
10:34:46 11 for any costs associated with the revenue that she's
10:34:50 12 modeling.

10:34:51 13 She's tried to model the time value of that revenue,
10:34:54 14 but that revenue isn't available 100 percent to Akoustis
10:35:00 15 because they spent more money making and selling the
10:35:03 16 product than they actually earned in selling it to date.

10:35:10 17 **Q.** By the way, I neglected to ask you, Ms. Irwin, in
10:35:14 18 your experience, have -- as a testifying expert, have you
10:35:19 19 ever analyzed the head start benefit in one or more cases
10:35:25 20 before?

10:35:25 21 **A.** Yes. Many. Over the course of 30 years, it's
10:35:29 22 probably 20 to 25 times, and I've testified about the head
10:35:32 23 start period probably eight times in the last 12 years or
10:35:38 24 so.

10:35:39 25 **Q.** Okay. And I understand that you referred to, in your

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benefit any of the types of costs that you just mentioned?

A. No.

Q. So what did she do?

A. She simply looks at revenue and measures the time.

What she calls the "time value of money." The difference between receiving the revenue and at the time that they actually did, and then modeling it 55 months later, and then taking the difference between those two present values and calling that the benefit that Akoustis received.

But as we'll see, they haven't made any money off the gross profit level. So there's been no benefit under this model or methodology.

Q. In your opinion, is relying on revenue alone to calculate the head-start benefit consistent with the principles governing your area of practice?

A. No. It's inconsistent with most basic financial principles that I taught at USC, and I'm sure Ms. Bennis studied as well.

Q. And so is calculating the head-start benefit based on revenue, even if it's discounted by the time value of money, an accepted practice at all?

A. No. Now, let me back up a bit. I don't disagree at all with the concept of time value of money. Again, that's a very basic concept in finance. It's the notion

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10:49:01 1 that a dollar today is worth more than a dollar in a year
10:49:05 2 or 55 months. But you have to understand what you have
10:49:10 3 available to actually invests and earn money on.

10:49:14 4 So another example is, say I inherit \$10,000 from my
10:49:20 5 grandmother. Thank you, Grandma. And if I just receive
10:49:23 6 that money, it's available for me to invest, to buy a
10:49:27 7 business, to help buy a house, whatever. But let's say
10:49:30 8 the IRS gets involved and says, No, no, no, you don't get
10:49:34 9 to keep the entire 10,000, you have to pay some taxes on
10:49:37 10 that.

10:49:38 11 Say the taxes came out to be about 3,000 -- I'm from
10:49:39 12 California, so it would probably be higher -- but the
10:49:42 13 difference that I have is only \$7,000 that's available to
10:49:47 14 invest. The time value of money is only applicable to
10:49:50 15 what's available to invest. Revenue is not available to
10:49:55 16 invest.

10:49:55 17 Only after you pay for some sort of costs -- and you
10:49:56 18 can debate which level here is relevant, but you have to
10:50:00 19 at least pay for the cost of revenue or the cost of sales
10:50:04 20 before you determine anything that's available to invest.

10:50:07 21 **Q.** So hypothetically, let's assume that in the slide
10:50:11 22 that we have in front of us, Slide 8, that the cost of
10:50:16 23 revenue -- that the inflow and outflow were flipped, so
10:50:20 24 that would -- would that give you a gross profit of
10:50:25 25 \$3.1 million?

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10:50:27 1 **A.** Yes. So if the revenue was 30 million and the cost
10:50:31 2 of revenue was 27, you would have \$3 million available to
10:50:35 3 invest. The opposite is true here. So it actually costs,
10:50:38 4 at least in fiscal 2023 -- and that's been true for a lot
10:50:41 5 of years now -- the cost of revenue has exceeded the
10:50:45 6 actual revenue.

10:50:46 7 **Q.** So in that case, could you, then, apply the time
10:50:49 8 value of money principle that Ms. Bennis tried to use in
10:50:56 9 her calculation?

10:50:58 10 **A.** You could, but you would get a result that indicates
10:51:01 11 that this is not the right methodology. In other words,
10:51:03 12 you could get a nonsensical result.

10:51:06 13 **Q.** Okay. We'll talk about that in a second.

10:51:07 14 Are you aware of any texts or scholarly articles,
10:51:12 15 like the Sedona Conference journal, approving Ms. Bennis'
10:51:17 16 methodology of using only revenue to measure the
10:51:21 17 head-start benefit?

10:51:22 18 **A.** I am not.

10:51:25 19 **Q.** Do any texts or articles, of which you are aware,
10:51:29 20 discuss Ms. Bennis' revenue methodology at all?

10:51:35 21 **A.** Not as she's applied it here.

10:51:37 22 **Q.** What effect does not including costs have on
10:51:41 23 Ms. Bennis' estimate of the head-start benefit?

10:51:44 24 **A.** It makes it completely unreliable.

10:51:46 25 **Q.** Okay. And is that not offset by her use of the time

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10:51:56 1 value of money?

10:51:57 2 **A.** No. Again, the only amount that's available to earn,
10:51:59 3 you know -- and can receive any time value of money to
10:52:03 4 benefit from is the difference between the revenue you
10:52:06 5 bring in and the cost to make the -- to generate the
10:52:09 6 revenue. You have to include costs.

10:52:12 7 **Q.** Okay. So could one fix Ms. Bennis' estimate of head
10:52:18 8 start benefit by accounting for Akoustis' costs?

10:52:25 9 **A.** You could. Again, it generates a nonsensical result,
10:52:30 10 and I can demonstrate how that's -- how it comes out. But
10:52:34 11 the fundamental reason why you can't really fix it is the
10:52:38 12 head start methodology is intended for a defendant that
10:52:43 13 has accelerated profit. All of the examples that I read
10:52:47 14 to you from the Sedona conference materials talked about
10:52:50 15 accelerated profits. We could debate whether it's gross
10:52:54 16 profit or operating income, but they all talk about
10:52:57 17 profit. And in this case, as you can see, Akoustis at the
10:53:01 18 gross profit level -- so that's just sales minus the cost
10:53:04 19 of sales -- had a very small profit for several years, and
10:53:07 20 then massive losses at the gross profit level through
10:53:12 21 2023.

10:53:13 22 And the cumulative gross profit losses for all of
10:53:17 23 that period of time is over \$10 million. So there's not
10:53:21 24 hundreds of millions or \$60 million to invest; there's
10:53:26 25 negative \$10 million to invest at the gross profit level.

Irwin - Direct

10:53:31 1 Q. Does a company benefit from a head start on losses?

10:53:35 2 A. No, not under this methodology. The math doesn't
10:53:40 3 work.

10:53:40 4 Q. Okay. So what impact would including costs have on
10:53:44 5 Ms. Bennis' estimate?

10:53:45 6 A. So if you go to the next slide, what you will see is,
10:53:48 7 so I've laid out in Ms. Bennis' methodology that generated
10:53:53 8 her \$66 million, and the only thing I changed was, instead
10:53:57 9 of revenue, I used the gross profit numbers that you saw
10:54:01 10 on the previous slide and dropped them into her model. As
10:54:05 11 a result, in the actual world, Akoustis has lost
10:54:09 12 \$21 million in value. And in the delayed, they've lost
10:54:16 13 \$11 million. This means that they are worse off in the
10:54:19 14 actual world where they allegedly misappropriated than in
10:54:25 15 the world where they would have been delayed.

10:54:27 16 In other words, it would have benefited Akoustis to
10:54:29 17 allegedly misappropriate. And that's -- so I can't assume
10:54:32 18 damages and then have a negative damages number. That's
10:54:36 19 just nonsense. What that tells me as a practitioner is
10:54:38 20 that this is not the right methodology, the head start
10:54:41 21 method doesn't fit the facts in this case because there is
10:54:44 22 no benefit to any acceleration in losses.

10:54:48 23 Q. Okay. So let me back up. We originally started by
10:54:54 24 you identifying three issues: Ignoring costs related to
10:54:59 25 the production of BAW filters. The second one was